

Revision of O*NET Data Collection Instruments

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Addendum
*Revisions to O*NET Data Collection Instruments*
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Below is information regarding important updates to the project described in the report, *Revisions to O*NET Data Collection Instruments, Revised Version, October 2000*.

- G** This report contains information based on O*NET 98 and refers to information presented in O*NET 98. Since the publication of this report, the O*NET 3.0 database has been developed. The major difference between this database and the O*NET 98 database is its compatibility with the 1998 Standard Occupational Classification (SOC) System.¹ By making O*NET 3.0 compatible with the SOC system, the O*NET 3.0 database contains 974 occupations. O*NET 98 contains 1,122 occupations. Note, the Office of Management and Budget has mandated that all federal agencies' occupational classification systems be compatible with the 1998 SOC system.
- G** Specific modifications made to prototype data collection instruments that resulted in the updated questionnaires² are described in this report. The modified questionnaires were used in the O*NET Pretest Data Collection, and the same version will be used in future O*NET data collection efforts. However, it is important to note that the modifications to the questionnaires described in this report are not yet reflected in the database structure of O*NET 3.0. These questionnaire modifications will be incorporated into the next version of O*NET 3.0.
- G** O*NET 3.0 and O*NET OnLine, a web-based application that allows users to view and use the O*NET 3.0 database, can be accessed via the website, www.onetcenter.org

¹ United States Department of Labor, Bureau of Labor Statistics. (1999). *Revising the Standard Occupational Classification System*: Washington, DC: Author.

² The *O*NET Data Collection Program* currently uses five questionnaires in data collection: Skills, Knowledges (which includes education, training and work style questions), Abilities, Generalized Work Activities, and Work Context.

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Executive Summary

The *Occupational Information Network (O*NET)* is a database describing the characteristics of workers and jobs. It contains information on 1,122 occupations. O*NET was designed to replace the *Dictionary of Occupational Titles (DOT)*. O*NET is the most comprehensive standard source of occupational information in the United States. O*NET will be at the center of an extensive network of occupational information used by a wide range of audiences, from individuals making career decisions, to public agencies and schools making training investment decisions, to employers making job structure and hiring decisions. O*NET will also be widely used for administration of federal programs, including employment, training and education programs of the Departments of Labor, Health and Human Services, and Education.

Information presented in O*NET is based on the O*NET Content Model (Peterson, Mumford, Borman, Jeanneret, & Fleishman, 1995). This content model covers six domains of job characteristics. Jobs can be described either by the demands placed on people doing the work (domains of Worker Characteristics, Worker Requirements, and Experience Requirements) or by the work being done (domains of Occupational Requirements, Job-Specific Requirements and Occupation characteristics). The data in O*NET describe each domain exhaustively. The data will be kept up-to-date by means of an extensive data collection program.

The O*NET data collection program uses a two-stage sample design, including a statistical sample of businesses and a sample of workers in selected occupations within the sampled businesses. These ‘incumbent workers’ are asked to complete the survey instruments which measure the job characteristics specified by the Content Model. The data collection program is a continuing activity to develop and maintain current job-related information. This information is crucial to the validity of O*NET.

In the initial field tests using prototype versions of the O*NET survey instruments, the response rates for incumbent workers were low. The O*NET Instrument Revision Working Group was convened to revise the prototype instruments in ways that would lead to improved survey response rates. The research reported here is an implementation of a philosophy of continuous improvement that has been adopted as an operational philosophy for all subsequent O*NET development.

This report is a summary of the methods and results of the Instrument Revision Working Group's modification of the prototype instruments. It describes:

- the cognitive methods used to evaluate the prototype instruments;
- the results of the cognitive evaluation of the prototype questionnaires. (The evaluation indicated that the questions were difficult to understand and answer, placing an unnecessary burden on respondents);
- the modifications in the design of those questionnaires made as a result of the evaluations. (The design modifications simplified the instructions and questionnaire items without altering their content);
- the revisions in the content of the prototype questionnaires. (The content revisions involve 1) reductions in the number of items, 2) reductions in the number of response scales per item, and 3) minor wording changes to specific variable definitions);
- the results of a brief laboratory pilot test of the revised instruments.

Chapter 1. Introduction

Historical Background

The *Occupational Information Network (O*NET)* is a new product, developed by the U.S. Department of Labor. It is designed to replace the obsolete *Dictionary of Occupational Titles (DOT)* and to provide additional information not previously available in the *DOT*. O*NET uses a new data structure, the Content Model, to conceptually organize occupational information, moving from the *DOT*'s over 12,000 occupations to a more manageable 974 occupations based on the *1998 Standard Occupational Classification*. O*NET's features, coupled with these compatibilities, provide a "common language" of standardized descriptors and measures about jobs for use by a variety of audiences interested in occupational information.

O*NET Content Model

The substantive domains that are to comprise the O*NET data base of occupational information were specified by the Advisory Panel for the Dictionary of Occupational Titles (APDOT, 1993). APDOT envisioned O*NET as comprehensively integrating all major types of occupational information in a systematic way. Thus, O*NET is a comprehensive database system for collecting, organizing, describing and disseminating information on job characteristics and worker attributes.

The O*NET system is based upon the O*NET Content Model (Peterson, Mumford, Borman, Jeanneret, & Fleishman, 1995). The Content Model underlying O*NET organizes job-related information into four types of cross-job descriptors:

- ***Worker Characteristics*** — abilities, interests, and work styles;
- ***Worker Requirements*** — basic skills, cross-functional skills, general knowledge and education;
- ***Experience Requirements*** — licensing, experience and training, and job-entry requirements for basic and cross-functional skills;
- ***Occupational Requirements*** — generalized work activities, organizational context, and work context.

Each domain of the Content Model contains a hierarchical structure used to group specific detailed information. For example, the *Worker Characteristics* domain contains three types of information: *Abilities*, *Interests* and *Work Styles*. *Abilities*, in turn, contains four types of

abilities, *Cognitive, Psychomotor, Physical and Sensory*. Each of these types of abilities contains further levels of detail. For example, Psychomotor abilities include *Fine Manipulative, Control Movement, and Reaction Time and Speed* abilities. Finally, Fine Manipulative Psychomotor Abilities contain three specific descriptors: *Arm-Hand Steadiness, Manual Dexterity, and Finger Dexterity*.

In addition to cross-job descriptors, the Content Model also includes:

- ***Occupation Specific Information*** — tasks that are particular to a specific occupation; and
- ***Labor Market Characteristics*** — such as pay and job openings.

The instruments used for O*NET development are designed to measure the various domains of the Content Model¹. Information for the model domains is collected by a series of questionnaires completed by job incumbents. Respondents to O*NET data collections are guided through a series of questionnaires designed to solicit ratings of jobs or workers represented by each element:

- ***Worker Characteristics*** are measured using the *Abilities* and *Work Styles* questionnaires;
- ***Worker Requirements*** are measured using the *Skills* and *Knowledges* questionnaires;
- ***Experience Requirements*** are measured using the *Training, Education, Licensure, and Experience* questionnaire;
- ***Occupational Requirements*** are measured using the *Generalized Work Activities* questionnaire, and the *Work Context* questionnaire;
- ***Occupation Specific Requirements*** are not measured using questionnaires, but are measured by job analyst observations, job holder and supervisor descriptions elicited in group discussions, and task inventories;
- ***Occupation Characteristics*** are also not measured using questionnaires, but are provided by linkages to other databases produced by the Bureau of Labor Statistics and its cooperating state employment security agencies, the National Occupational Information Coordinating Committee, the U.S. Department of Education, the Office of Personnel Management, and others.

¹ Due to federal government data collection authority, response burden concerns, resource constraints, etc., not all portions of the Content Model (or associated questionnaires) are to be applied within every fiscal year.

Prototype O*NET Measures

The prototype O*NET questionnaires used several types of rating scales, depending on the element being measured. Most were scales for measuring Frequency, Importance, Level, or Agreement. For example, the prototype Generalized Work Activities Questionnaire elements were rated according to Level, Importance, and Frequency.

The prototype instruments have been used in three data collections:

- In the first, pilot test versions of the instruments were administered across the country to small opportunity samples at five employment service offices. Respondents were locally employed adults who volunteered to attend the sessions. The aggregate sample consisted of 209 people sampled from 20 occupations. A structured questionnaire with some open ended items was used to solicit evaluative comments about the instruments.
- In the second data collection, a mail-out survey obtained ratings on the O*NET descriptors for all domains from 15 job incumbents in each of 80 occupations. It was this survey that yielded a low return rate.
- In the third, data collection consisted of having occupational analysts rate each of the O*NET occupations on selected O*NET variables. Each occupation was rated on the basis of a definition and written task description that consisted of a representative set of job tasks (taken from the DOT codes that comprise that occupational unit).

The prototype O*NET measuring instruments were developed over a relatively short period of time. Each instrument was produced by a domain research team that surveyed the available literature and generated a taxonomy of variables that, in their judgment, best represented and synthesized the possibilities presented by the literature. The previous literature varied from extensive, in the case of “abilities” (e.g., Fleishman, 1992; Carroll, 1993), to minimal, in the case of “skills” (e.g., see Lubinski and Dawis, 1992). The decision to include and characterize constructs or variables involved some subjective decisions based on these literatures (Peterson et al, 1995; Mumford, Sager, Childs, & Whetzel, 1995; Peterson, 1997). In other words, in many instances the domain teams had to make a reasoned choice about what to include or not to include, given a lack of clear guidance from the literature.

It is for such reasons that neither the prototype model, nor the instruments developed to measure it, was viewed as the final answer. The expectation by the prototype development team was that there would be several stages of revision and improvement (Peterson et al., 1995). Such revisions and improvements would be driven by technical and scientific advances, by economic and evolutionary forces in the labor market, and by taking advantage of opportunities for improvement that arise as part of the operation of the O*NET system. Consequently, continuous improvement has been adopted as an operational philosophy for all subsequent O*NET development.

Revision of the Prototype Measures

Based on the initial assessment of the prototype instruments, and on the philosophy of continuous improvement for O*NET, the present efforts to revise the instruments were undertaken. The remainder of this report describes the methods and results of these revisions. **Chapter 2** describes the cognitive methods used to evaluate the prototype instruments. **Chapter 3** reports the results of the cognitive evaluation of the prototype questionnaires and describes the modifications in the design of those questionnaires made as a result. The evaluation indicated that the questions were difficult to understand and answer, placing an unnecessary burden on respondents. The design modifications simplified the instructions and questionnaire items without altering their content. **Chapter 4** describes the revisions in the content of the prototype questionnaires. Those revisions involve 1) reductions in the number of items, 2) reductions in the number of response scales per item, and 3) minor wording changes to specific variable definitions. This report concludes by providing the results of a brief laboratory pilot test of the revised instruments in **Chapter 5**.

Chapter 2. Methods for Evaluation of Prototype Instruments

Introduction

On April 30, 1998, the Instrument Revision Working Group, (See **Table 1**) was convened with the charge of identifying possible changes in the O*NET instrumentation. This group was charged with making changes that would reduce the respondent burden (thereby increasing response rates) while keeping intact the Content Model. These considerations encouraged thorough appraisal, candid commentary, and, where appropriate, pointed criticism of previously planned practices. In order to make decisions on what changes to consider, several alternative forms of evaluation were utilized. In this chapter we describe the methodological approach used to evaluate the prototype instruments.

Table 1. Instrument Revision Working Group

<u>Member</u>	<u>Affiliation</u>
Rodney McCloy	Human Resources Research Organization
John P. Campbell	Human Resources Research Organization
Michael Hubbard	Research Triangle Institute
John Nottingham	National Center for O*NET Development
Phil Lewis	National Center for O*NET Development
David Rivkin	National Center for O*NET Development
Jonathan Levine	National Center for O*NET Development

Cognitive Methods: Overview

The O*NET prototype instruments were evaluated using cognitive methods for questionnaire evaluation and design. Based on theory and methods of cognitive psychology, the cognitive approach to questionnaire design and evaluation has generated much research (Jobe & Mingay, 1991; Jobe, Tourangeau, & Smith, 1993; Lessler & Sirken, 1985). In the Federal Statistical System, the approach is applied at the National Center for Health Statistics (Royston, Bercini, Sirken & Mingay, 1986; Willis, Royston, & Bercini, 1991), at the Bureau of Labor Statistics (Dippo, 1989; Esposito & Hess, 1992) and at the Census Bureau (Campanelli, Rothgeb, & Martin, 1989; Campanelli, Martin, & Rothgeb, 1991; De Maio, Martin, & Sigman, 1987). A review of the work done at these agencies is contained in Jobe and Mingay (1991). The overall goal of cognitive evaluation is to identify those aspects of survey questions that make them easier to understand and answer.

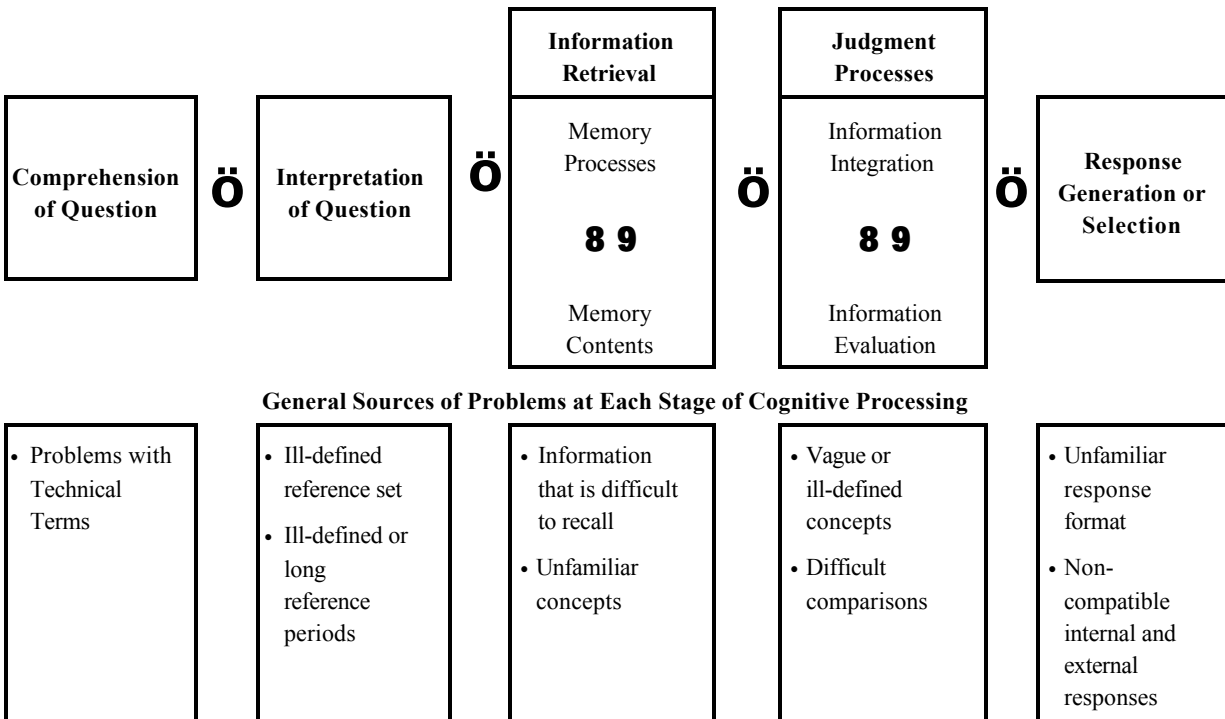
Cognitive methods focus on the questionnaire instrument, rather than on the entire survey process, and they can be applied at several points in the design process. According to Forsyth and Lessler (1991), cognitive methods can be categorized into four general types: *expert evaluation* (e.g., expert analysis or cognitive forms appraisal); *expanded interviews* (e.g., concurrent and retrospective think-aloud interviews); *targeted methods* (e.g., paraphrasing or rating tasks); and *group methods* (e.g., focus groups or group interviews). Each method has its strengths and limitations. In terms of instrument development, however, all cognitive laboratory methods share the advantage of providing information about the likely performance of the survey instrument in the field without incurring the high cost of full-scale field testing. All but the targeted methods approach were used in the evaluation of the prototype O*NET data collection instruments.

Expert Evaluation

In *expert evaluation*, a knowledgeable researcher reviews a questionnaire to gather an understanding of the response task and to note potential problems. The questionnaire's characteristics are described, including procedures for moving through the questionnaire, type of respondents, and the tasks required of the respondents. Lists are made of problems that respondents might have and types of mistakes they might make. The researcher then classifies the observations to get a general understanding of the questionnaire and specific points where difficulties may occur.

Cognitive forms appraisal (Forsyth, Lessler, & Hubbard, 1992) is one method of expert evaluation in which the appraiser systematically analyzes questions using a specific model of the response process. The survey response model used in the cognitive forms appraisal was one developed by Forsyth and Lessler (1991) (see also Forsyth, Hubbard & Lessler, 1990). It is based on cognitive models of the survey response process developed by survey methodologists (Cannell, Miller, and Oksenberg, 1981; Tourangeau & Raskinski, 1988; Willis, Royston, & Bercini, 1991) and similar to more general cognitive information processing models (Wyer & Srull, 1989; Wyer & Radvansky, 1999). **Figure 1** is a schematic of the response process model. **Figure 1** also indicates the kinds of difficulties with survey questions that are often identified by a cognitive forms appraisal.

Figure 1. Cognitive Model of the Survey Response Process



To illustrate the stages of the information processing, consider **Figure 2** which displays a typical questionnaire item from the prototype O*NET Knowledges questionnaire. Note that there are two questions asked on the page: the Level question in the top half of the box and the Importance question in the bottom half.

- First, the respondent must **comprehend** the question to understand what is being asked. With regard to the level question, the respondent must first find the question on the page, and understand that it asks about the level of administration and management knowledge, as defined, needed to perform his/her job. For this item, the layout of the page, the definition of the knowledge area, and the boxed descriptions of high and low levels of the knowledge area may be difficult to comprehend.
- The respondent must then **interpret** the question (form a mental representation of the target) that he or she must evaluate and decide about range of relevant concepts (the *reference set*) of the question. For the Level question, the respondent must decide how the definition of administration and management knowledge applies to his/her job in general. If the worker is, say, a beautician, she may find this step difficult, given the definition of administration and management provided.

Figure 2. Example Prototype Questionnaire Item (from Knowledges questionnaire)

1. Administration and Management

Knowledge of principles and processes involved in business and organizational planning, coordination, and execution. This includes strategic planning, resource allocation, manpower modeling, leadership techniques, and production methods

<u>Level</u>				
What level of this knowledge is needed to perform this job?				
Requires knowledge of high-level business and management principles and processes such as those required in large, multi-national organizations with diversified products or services	ô ó ò ñ ð ã î	2 2 2 2 2 2	Managing a \$10 million company Administering a large retirement and nursing care facility Monitoring progress of a project to ensure timely completion. Planning an effective staff meeting. Signing a pay voucher	
	NR Not relevant at all for performance of this job			
	Requires knowledge of basic business and management principles and processes such as those required in small businesses with a single product or service.			
	<u>Importance</u>			
	How important is this knowledge to performance on this job?			
	Not Important î	Somewhat Important ï	Important ð	Very Important ñ

- The respondent then **retrieves information from memory** (e.g., recalls relevant instances of behavior or relevant opinions), while taking into account the relevant time frame (the *reference period*: e.g., what was done last week or last year) or else reconstruct the behavior using general knowledge or salient information. In the Level question on Figure 2, our beautician must try to remember instances of how she applied knowledge of administration and management in her work - perhaps coordinating work schedules among fellow employees.
- Next, the respondent must make a **judgment** to determine whether the remembered instances of behavior are reasonably representative and if they are relevant to their interpretation of the question. Our beautician must decide if work scheduling is really part of her job and if it counts as being administration and management. If not, she must try to recall other, more typical activities that may fall into the general category.
- Once a private (internal) judgment is formed in the respondent's mind, he or she must **respond** by communicating an answer to the researcher. To do so, he or she must fit the judgments to the response format provided as part of the question. The respondent may also edit the response, before communicating it, to make it socially desirable or appropriate to the situation. Our beautician must figure out where her coordination of work schedules falls on the dimension anchored at Managing a \$10 million company (6 out of a possible 7) and signing a pay voucher (1 out of 7), and decide if she's being overly boastful or self-effacing by indicating that her administrative activities represent a 3 on that scale.

The model analyzes the answering of survey questions as a series of cognitive tasks. Although depicted as a linear, sequential process, answering a question (especially one that is hard-to-understand or hard-to-answer) can go through multiple cycles of this sequential process until a suitable answer is produced.

Expanded Interviews

Expanded interview methods use survey respondents, rather than experts, to provide information about what it's like to answer the survey questions. Expanded interview methods include concurrent and retrospective think-aloud interviews. In these interviews, the respondent verbalizes his or her thoughts while answering the question (concurrent) or recalls how he or she answered the question (retrospective). In these interviews, the interviewer may ask follow-up probe questions to explicate respondent answers, or provide memory cue tasks to encourage the respondent to encourage more complete recall. Often the follow-up probe questions and memory cue tasks are developed to verify the reality of a potential problem identified by expert evaluation. There is often a debriefing of the respondent at the end of the interview in which he or she discusses general impressions of the survey questions and what strategies were used in answering them.

O*NET data collection instruments were evaluated by Lesgold, Resnick, Katz, and Hwang at the Learning Research and Development Center, University of Pittsburgh. Their think-aloud interview protocols were directed toward respondents' ability to make valid ratings of skill levels and work/worker characteristics needed for a given job. The researchers were particularly interested in how respondents interpreted descriptions of skill and work activities, and how they used the descriptive anchors provided in the questions to make judgments about their own jobs.

Group Methods

Group methods of instrument assessment include focus groups and group interviews. A *focus group* is a small group of individuals who are convened to discuss a specific topic under the direction of a moderator who promotes interaction and assures that the discussion remains focused on the topic. A focus group produces qualitative data that provides insights into the attitudes, perceptions and opinions of its participants. These results are solicited through semi-structured group discussions during which group participants influence each other – as they may in real life. In a *group interview*, an interviewer takes a more directive role in leading discussion of the topic of interest, rather than having group members talk freely.

The O*NET data collection instruments were reviewed by a focus group of employees of Research Triangle Institute. They were provided with copies of the O*NET questions in their original format and also the same question content presented in two alternative formats. Both questions and instructions were modified in the alternative formats. After completing the instruments, a group discussion was held focusing on usability of the instruments in general, with particular emphasis on the question formatting. The group was moderated by Dr. Michael Hubbard and Ms. Connie Hobbs.

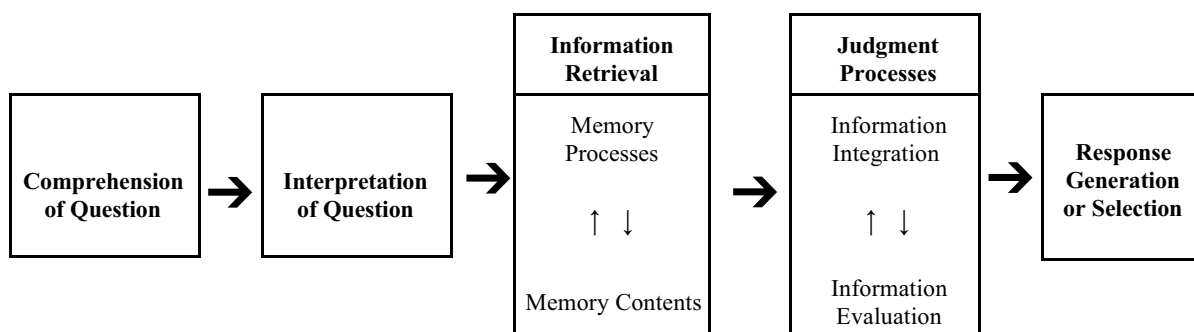
Chapter 3. Evaluation of Prototype Instruments and Modification of Questionnaire Design

Overall Evaluation of Prototype Questionnaires

All three sources of cognitive evaluation information (expert appraisal, expanded interviews and group methods) agreed in suggesting that there were potentially serious problems with the prototype O*NET instruments. They all pointed to the possibility that at least some respondents could find the questions hard to understand, interpret, and answer. This difficulty might lead respondents to refuse to complete the questionnaires, resulting in low response rates and other consequent problems.

The cognitive appraisal of the prototype instruments in general suggested that there were problems at all stages of the question-answering process. *Figure 3* includes a summary of the kinds of problems identified by the cognitive appraisal. Discussion of specific instruments, below, will provide more detail regarding these problems.

Figure 3. Problems with O*NET Questions Identified by Cognitive Process Model



Difficulties with O'NET Questions

• Undefined technical terms • Vague or Ambiguous technical terms • Confusing examples • Long sentences with polysyllabic words	• Many abstract concepts • Assumes knowledge of conceptual domains of interest • Vague or abstract reference set • Unspecified reference period	• Unfamiliar level of abstraction for usual activities • Novel concepts related to job characteristics	• Difficulty combining abstract concepts with unfamiliar views of own behavior	• Difficulty locating answer on unfamiliar response scales
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General Evaluation of Box-Format questionnaires

The Abilities, Generalized Work Activities, Knowledge, Skills, and Work Context questionnaires, (the largest of the prototype O*NET questionnaires) share a common format that poses a common set of challenges to respondents who complete them. They all have a similar appearance on the page: concepts are defined at the top of the page and questions are placed inside boxes. Hence they are referred to here as *box-format questions*. An example of a box-format question can be seen in **Figure 2**, the example questionnaire in **Chapter 2**.

All box-formatted items call for respondents to understand what is often an unfamiliar, novel and often highly abstract concept. They must then evaluate their jobs (not just themselves or their behavior) and make a series of judgments as to the applicability of the concept to their job, the level of the concept embodied in their job, the importance of the concept to their job and, sometimes, another judgment (depending on which questionnaire). To do this once is a challenging cognitive task and respondents may be called on to do this up to 50 times in the process of completing the survey.

The task of answering such an item requires a fair level of verbal ability (to understand the words presented), cognitive sophistication (to conceptualize these novel concepts), and abstract thinking (to generalize from their own experiences to make judgments about people who they imagine might do their job). Respondents must also be sufficiently familiar with questionnaires to easily project their mental representation onto a rating scale that is itself complex and difficult to understand. They must also be motivated to do this repeatedly on their own time. According to the cognitive appraisal, the box-format questionnaires had several problems, both with instructions associated with them and with the questions themselves.

Evaluation of Instructions for Answering Box-Format Questions. The instructions in all of the box-format questions consist of three or four pages: one page (or two, depending on the number of boxes per page in the questionnaire) consists of written descriptions of what the scales are and how to use them to answer questions. The remaining pages are examples of questionnaires “completed by an employee in a particular/different job”. The expert evaluator of the instructions found that they were written at too high a reading level: that they were overly long and unnecessarily abstract. The focus group comments were in agreement with this evaluation. Focus group participants thought the instructions had many problems: They were too long (should be only 1 page, perhaps with an additional page for an example), wordy, vague, and hard to read. Focus group members thought these problems would tend to discourage many people faced with making 30-50 more ratings as their task.

Evaluation of Box-Format Questionnaire Items. The formal cognitive appraisal of these questions identified the following areas in need of improvement:

- **QUESTION STRUCTURE:** Questions are complex in the way they are arrayed on the page. The pages in the Generalized Work Activities, Skills, Abilities, Work Styles, and Knowledges questionnaires present a challenging visual array, the complexity of which reflects the complexity of the questions. There are boxes within boxes, circled numbers, arrows, changes in type font, horizontal scales, vertical scales, italics, and bolding on each page. While respondents may become familiar with such pages, it requires a certain level of sophistication with forms and questionnaires — a level that may not be present in respondents with low levels of education (or very low levels of experience with such measures).
- **QUESTION CONTENT:** The questions call on the respondent to make judgements that are unfamiliar. Respondents are asked to rate their job in the abstract, rather than what they actually do on the job. This makes the questions more burdensome. They would be easier to answer if respondents are asked about what *they* do. This complicates the interpretation and memory tasks because the respondents must filter their memories and judge whether they are part of the job (rather than their own idiosyncratic behaviors). Many workers may have never distinguished between their personal job experiences and their job as such.
- **TECHNICAL TERMS:** There are many unfamiliar terms in these questions. Although they are defined and examples of them are provided, they are still often insufficiently defined or else they remain vague or ambiguous, especially for people who are unfamiliar with the relevant concepts. The fact that questions may require the respondent to think of their jobs in novel or unfamiliar ways, makes understanding/application of the terms more difficult.
- **TECHNICAL TERMS:** Most of the questions refer to high level abstractions of everyday activities. This makes it hard to (1) understand what the question is asking about and (2) retrieve and combine information to formulate an answer. This can result in unreliable and invalid answers. *It is likely that high levels of education - higher than high school - are necessary, due to the extensive vocabulary used in these instruments.*
- **REFERENCE SET:** Workers may not routinely think of jobs in terms of the level, importance, and job entry requirements of job-related skills.
- **REFERENCE SET:** Each **Level** question includes labels for interpreting the extreme values on the scale. For two questionnaires (Skills and Work Styles), the scales are anchored simply by HIGH and LOW. These scales are comparatively easy to rate because ratings of the level are reflections of the amount of the concept in question.

However, the concepts themselves, (e.g., *Integrity, Evaluating Information For Compliance to Standards* or *Synthesis/Reorganization*) may vary in how understandable they are. For the remaining questionnaires, the ends of the scales are identified by labels containing many words (as many as 40) and often containing numerous long, polysyllabic abstract or technical words. Thus the readability of these dimensional anchors may be poor. This makes it difficult for respondents with low levels of education to use the scales.

- **REFERENCE SET:** Each level question also includes examples of the concept to which the question refers. Some of these examples may be beyond the range of experience of many respondents. Even though they may be intended only to convey a general sense of the level of the concept, they can contribute to a sense that the questions are difficult, which can decrease effort on the questionnaire or completely discourage its completion.
- **REFERENCE PERIOD:** There is no explicit time period mentioned in the questions. While most may consider the job as it is now, others could also consider how the job was previously.
- **RESPONSE SELECTION:** Respondents are given a 7-point vertical scale for indicating the levels of the concepts, and 5-point horizontal scales for rating importance. They may have trouble alternating between the two scales.

The focus group comments were also critical of the box-format questionnaires. Focus group participants didn't like the (7-point vertical) scale, finding it hard to mark (it implied that marks between numbers were permissible, especially since example anchors pointed between numbers). They also didn't like numbers without verbal anchors. They found the descriptive anchors in the Level question box hard to understand. In cases where the definitions of high and low level attributes were provided in boxes within the question box (see Figure 2), focus group participants thought these definitions were hard to understand. Focus group participants sometimes found the right-side (example) anchors to be of limited utility in making a rating judgement about their job.

Focus group participants also didn't like the mixture of 5-point and 7-point scales and vertical and horizontal scales and thought that there should at most be 5 scale points on such complex rating scales (although substituting 5-point scales for 7-point scales might exacerbate the problem of example anchors pointing between numbers). They pointed out that the NR instruction (at the bottom of the box), though mentioned in the instructions, was soon forgotten or lost amid the visual complexity of the answer sheets.

General Modification of Box-Format Questionnaire Items.

Based on the evaluations of the instruments outlined above, extensive redesign of the prototype instruments was undertaken. Modifications were made that would ease the response burden on respondents without changing the content of the questions.² Changes in the questionnaire formatting combined (1) responses to the issues raised in the questionnaire evaluation with (2) the expert judgment of the instrumentation working group. Many different formats were considered for the box-format questions, before the final recommended version was adopted. Included here are descriptions of the final recommended questionnaire formats (and comparisons between them and the initial formats).

Modification of Box-Format Questionnaire Instructions. The modified instructions for the box-format questionnaires have been shortened to one page (except for the Knowledges Questionnaire, which requires two pages to instruct respondents how to answer three questions about each knowledge area, instead of one or two as is the case in the remaining box-format questionnaires). An attempt was made to keep the instructions to a minimum. **Figure 4** is an example modified instruction page: the instructions for making Skills ratings.

² However, the modified versions of the questionnaires were also changed for other reasons (to reduce instrument redundancy and to improve readability) which did alter the questionnaire content. These modifications are discussed in **Chapter 4**.

Figure 4. Example of Modified Instruction Page for Box-Format Questionnaires

Instructions for Making Skills Ratings

These questions are about work-related skills. A **Skill** is the ability to perform a task well. It is usually developed over time through training or experience. A skill can be used to do work in many jobs or it can be used in learning. You will be asked about a series of different skills and how they relate to *your current job* - that is, the job you hold now.

Each skill in this questionnaire is named and defined.

For example:

Writing	Communicating effectively in writing as appropriate for the needs of the audience.
----------------	---

You are then asked to answer two questions about each skill:

A *How important is the skill to the performance of your current job?*

For example:

How <u>important</u> is WRITING to the performance of your current job?				
Not Important*	Somewhat Important	Important	Very Important	Extremely Important
1	2	3	4	5

Mark your answer by putting an **X** through the number that represents your answer.

Do not mark on the line between the numbers.

***If you rate the skill as Not Important to the performance of your job, mark the one **1** then skip over question B and proceed to the next skill.**

B *What level of the skill is needed to perform your current job?*

To help you understand what we mean by **level**, we provide you with examples of job-related activities at different levels. For example:

In the first paragraph, the content of the questionnaire is named and defined. Then the respondent's task is described. The remainder of the instructions describe the parts of each item and how to go about answering the questions accompanying each item. Respondents are explicitly told to mark the numbers and not to mark the line connecting them.

Accompanying each description is an example of how to answer the particular type of question being described. Because the example immediately follows the description of the question, there is less demand on the respondent to remember the description of the question before being shown how to answer it. In the original versions of the instructions, longer descriptions of all question types preceded examples of how any question should be answered. If the respondent considers the characteristic as not at all important to their job, they were instructed to mark the *Not Important* box on the scale and to skip subsequent questions about that characteristic. This instruction is explained on the instructions page and repeated for every characteristic for the revised box-format questions. In the prototype instruments, the instruction to skip subsequent questions (for characteristics seen as unimportant to the respondent's job) was provided only in the instructions and not thereafter.

Modification of Box-Format Questionnaire Items. There were a number of changes in the formatting of the box-format questionnaires. Decisions about these changes were made in a series of meetings and electronic conferences in which various alternative formats were presented and discussed.

- The boxes surrounding each question, and the labels for those boxes (i.e., Level, Importance, Frequency) were eliminated. This reduced the level of visual clutter of each page by removing unnecessary distractions from the questions. The convention of numbering the characteristics while distinguishing between questions about each characteristic was retained by labeling the Level question **A** and the Importance question **B**. The instructions distinguished between the two question types in a similar manner. All questions were left justified (instead of centered) to make it easier to locate the beginning of each question in the absence of the boxes.
- The name of the characteristic about which judgments are being made (e.g., the Skill, Generalized Work Activity, or Ability) was inserted into each IMPORTANCE and LEVEL question. Previously, these questions all asked about the importance or level of "this skill/activity/ability" without reminding the respondent what they were rating. This was done to reduce the memory burden on respondents in answering the survey questions.

Changes in the IMPORTANCE questions.

- The IMPORTANCE question was moved so that it would be the first question asked about each characteristic.

- The line connecting the numbers on the IMPORTANCE scale ended with the numbers **Î** and **Ô**, rather than extending beyond them as was done previously. This made the scale consistent with the LEVEL scale in this regard.
- The Not Relevant option became part of the IMPORTANCE question, represented at the extreme low-importance end of the scale. Thus, it was removed as an alternative for the LEVEL question.
- The instruction to skip subsequent LEVEL (and other questions, where appropriate) was repeated for each characteristic immediately following the IMPORTANCE question. This was done to increase the likelihood that respondents would appropriately skip unnecessary questions, thereby reducing respondent burden.

Changes in the LEVEL questions.

- The response scale was presented in a horizontal format, instead of its original orientation.
- For questions where descriptions of highest and lowest levels of the characteristic were indicated by long boxed definitions (such as in *Figure 1*) these were eliminated and the only label for the entire scale was the word *Highest Level* located below the **Ô** (the scale end point). This same formatting was applied to questionnaires (such as the *Skills Questionnaire*) where the ends of the level questions were labeled simply High and Low.
- The anchors providing examples of specific job-related activities were rounded to make them point to specific scale numbers, rather than between numbers as was done previously. Although this might compromise the apparent ratio level of measurement implicit in the precise placement of the arrows (which were based on results of statistical analysis from previous studies), the response task was made easier by emphasizing the use of numbers, instead of suggesting that marking between numbers was acceptable.³ This was reinforced by the instruction not to mark on the line between the numbers. However, the difference between the 7-point LEVEL and the 5-point IMPORTANCE scale was retained. This was done to keep data from future surveys consistent with current O*NET information.
- The number of specific job-related anchors was always three, as opposed to as many as nine in the previous questionnaire. These three anchors generally pointed at scale numbers 2, 4, and 6; although in some cases other numbers were appropriate. The three anchors were chosen to be those that were most concrete understandable among those available.

3

The psychometric properties of these anchors will be re-visited during O*NET's pre-test phase.

Overall Evaluation of Other Questionnaires

The prototype O*NET data collection instruments included other questionnaires, in addition to the box-format instruments. These include the *Background Information Questionnaire*; the *Training, Education, Licensure, and Experience Questionnaire*; and the *Work Context Questionnaire*.

These questionnaires were generally shorter than the box-format questionnaires (under 30 questions), with the exception of the *Work Context Questionnaire* which was among the longest questionnaires (107 questions). While not identical in format, they were subjected to cognitive appraisal, the results of which are discussed here. {The original set of instruments included the *Occupational Values Questionnaire*. However, this questionnaire will not be administered in the pretest}

Evaluation of Instructions for Other Questionnaires. Understanding how the questions were formatted posed less of a cognitive task than was the case for the box-format questions. As a result, the instructions were shorter. However, there were difficulties with the instructions. They were highly abstract and tended to be confusing. Instead of just telling respondents how to complete the questionnaires, they provided additional information about why the questions were being asked. This review found such information to be of questionable value. For instance:

- The instructions for the *Background Information Questionnaire* informs respondents that “In this questionnaire you will provide information about your background. ...Your responses to these questions will help to determine whether the goal of diversity is being achieved. ...Responding to these questions is optional; however, complete answers will make tracking the diversity of participants possible...” No further information was provided about what is meant by “your background” or “diversity”. The remainder of the instructions contained information that belongs in a statement of informed consent. Thus, since no instructions for completing the questionnaire were provided, simply having a cover with the name of the questionnaire, followed by the questions, would suffice.
- The instructions to the *Training, Education, Licensure, and Experience Questionnaire* were comparatively brief and to the point, although they could have been even shorter without loss of information.
- Likewise, the instructions to the *Work Context Questionnaire* were brief, although needlessly verbose. These instructions mention a reference period (which the other questionnaires fail to do), but they do not consistently guide all respondents to answer in the same manner. Subjects are left free to choose the reference period, so that answers of some subjects will be referring to a different time period than those of others. While this makes the instruction less burdensome or constraining, it makes data collected by different respondents difficult to compare (to the extent that they choose to consider different reference periods). The instructions also fail to tell the respondent how to answer the questions. Since most questions are answered using a continuous line accompanied by verbal anchors and circled numbers, it is not clear which of these should be used to answer the questions. Respondents may opt for making a mark on the line, which would make data entry difficult for those questions.

Evaluation of Questionnaire Items for Questionnaires. The formatting of the *Background Information* and the *Training, Education, Licensure, and Experience* questionnaires was not overly challenging. The only real formatting challenge among these instruments was that the *Training, Education, Licensure, and Experience* questionnaire used 6 different ways of formatting the questions within the same questionnaire. The formatting of the *Work Context Questionnaire* was more problematic. Some of the questions have dangling labels which identify what the question is about, but which are not part of the question itself. (e.g., 1: Formality of Communication, 27: Decision Latitude). Sometimes, these contain important information for interpreting the questions (as in Q17-21, where definitions of terminology are provided), but usually, they contribute nothing to the question being asked and are probably more of interest to researchers than respondents.

A more serious problem with the *Work Context Questionnaire* arises from the response scales used. While most questions use horizontal response scales, they change from 7-point to 5-point scales and back numerous times. There are also frequent changes in the scale anchors, which the respondent may or may not notice. For some questions, a 7-point scale is used without a verbal anchor for what each number represents, making it difficult to make distinctions among the unlabeled numbers.

Modification of Other Questionnaires

Modification of *Background Information Questionnaire* Instructions. The prototype instructions were reorganized so that the modified instructions contain two explanatory paragraphs and a new paragraph explaining how the questions should be answered. The first paragraph addressed issues of voluntariness and privacy. The second paragraph explained the purpose of the questionnaire and its importance. In the previous instructions, these themes were present in both introductory paragraphs, making it difficult to understand them. The previous version of the instructions had no instruction about how to answer the questions.

Modification of *Background Information Questionnaire* Items. The content of the prototype *Background Information Questionnaire* was modified by deleting questions about the respondent's name, organization, and other descriptors of his or her work. Formatting modifications corresponded to the modified content. Instead, the respondent was asked to write in his or her job title and to list how long they worked at their job. The remaining questions requested demographic information: year of birth, gender, Hispanic ethnicity, race, handicap and disability status. All questions would be answered by checking boxes or by writing in a number (for year of birth) or writing on a blank line (for job title, other Hispanic group, other race or specify Indian tribe).

Modification of the *Training, Education, Licensure, and Experience Questionnaire* Instructions. The modified instructions simply said what the questionnaire was about and told respondents how to respond (put an **X** in the box). Since the questionnaire was reformatted so that responding to all questions had the same response format, it was easier to tell respondents what to do. Previously, the variety of response formats made it necessary to tell respondents to “mark” the answers without saying how.

Modification of the *Training, Education, Licensure, and Experience Questionnaire* Items. The content of the prototype *Training, Education, Licensure, and Experience* questionnaire was modified by deleting questions about Required Instructional Programs, Level of Education in Specific Subject Areas, and Licensure, Certification, and Registration. It was renamed the *Education and Experience Questionnaire*. The question about required level of education was reformatted so that it can be answered by putting an ‘**X**’ in a box — instead on a short line before each answer option (making it consistent with marking other instruments). The experience questions were reformatted so that the set of answer options was repeated after each of the four questions. In the previous version, instructions for how to answer the questions were placed above the questions and the respondent would have to shift his or her attention repeatedly between the question and the instructions.

Modification of the *Work Context Questionnaire* Instructions. The modifications to the instructions for the *Work Context Questionnaire* involved a slight shortening of the wording. The previous version instructed respondents to consider the appropriate time frame when answering the questions. However, the wording of that instruction left so much to the discretion of the respondent that it was easier to simply eliminate it. The new instructions provided an example of a question and demonstrated how to answer it.

Modification of the *Work Context Questionnaire* Items. Modifications of the format of the *Work Context Questionnaire* were as follows:

- Questionnaire items were renumbered so that a single-level numbering scheme (all questions numbered 1, 2, etc.) was employed, rather than a two-level scheme (Sections of the questionnaire given numbers and questions within each section having lower-case letters: a, b, etc.). This involved removing the section head detached from any question (such headings are often overlooked or ignored) and making each question a unified whole that is easier to understand and think about (without searching the questionnaire for scattered pieces of the question).
- All questions were answered using horizontal 5-point scales, instead of shifting between 5-point, 6-point, 7-point, and 8-point scales as was done in the prototype questionnaire. Every scale point in the modified questionnaire had an accompanying verbal

- anchor instead of the prototype scales, some of which had as many as four of seven scale points unanchored. The only exceptions were the last two questions which used horizontal 3-point scales. These same questions were answered with the more traditional multiple choice formatting used in the previous version of the questionnaire.

This chapter presented the cognitive evaluation all of the prototype questionnaires and how they were modified to address the shortcomings identified by the evaluation. The evaluation indicated that the questions were difficult to understand and answer, placing an undue burden on respondents. This, in turn, could lead respondents to them give up because the task is too difficult or out of frustration, resulting in low response rates. The modifications suggested by the instrumentation working group were designed to alleviate the difficulties by simplifying the instructions and questionnaire items without altering their content. In **Chapter 4**, we will describe other modifications of the prototype questionnaires that did involve changes in the content of the questionnaires.

Chapter 4. Revision of Instrument Content - Scale Redundancy And Reading Level

Introduction

O*NET Prototype Project Director, Norman Peterson, in a memorandum to the Department of Labor, dated 4/9/97, outlined a number of options for reducing burden and increasing return rate. Eliminating descriptors, combining individual descriptors into composites, eliminating scales, eliminating domains, and considering some domains as the necessary core (while data from other domains would be collected if the opportunity presented itself) were all described as possible options.

Of these, he argued that the best options involved: a) an elimination of items that have the poorest conceptual and/or empirical support and b) combining items that took advantage of conceptual and empirical redundancy. He argued that these options presented some potentially significant advantages as well. For example, the validity of the taxonomy could be enhanced if the changes resulted in a taxonomic set of descriptors that were more correspondent with current and previous research, and which presented respondents with constructs to rate that were conceptually more distinct and more meaningful.

Peterson also argued that burden could be reduced considerably if some of the response scales were eliminated and that such an option should be carefully considered. In general, he argued for keeping the level scale and opting to eliminate “frequency” and/or “importance,” depending on the existing data.

The changes described here are consistent with Peterson’s suggestions and priorities. The next sections will summarize the specific changes and the reasons for them. Further, the proposed changes were few in number and constitute only minimal alterations to the content of the questionnaires. No changes were made to the “Content Model,” no domains were eliminated, and no changes were made to the meaning of each domain. That is, the changes that were made to the O*NET questionnaires do not alter the a priori conceptual factor structure for each domain that guided the development of the O*NET prototype (as described in Peterson, 1995).

Minimal though they were, the combined effects of small reductions in the number of items, carefully considered eliminations of some response scales, and changes in format readability have significantly reduced the sheer size of the questionnaires, altered their formidable appearance, and significantly reduced the time it takes to complete them.

General Issues Regarding Changes in Instrument Content

The next sections summarize the changes that were made. Appendix A presents the changes (between prototype and current version) in variable definition and wordings for each questionnaire. Appendix B lists the O*NET data dictionary variables that will be included in the current version of the O*NET Survey. Appendix C lists all variables from the O*NET data dictionary and indicates (a) changes in their definition or formatting, and (b) the sources of the data in the O*NET database. Before describing the changes, several general issues should be noted:

- Some of the items in all the instruments and all of the items in some of the instruments (i.e. Abilities, Styles, Skills) ask for judgments about variables — variables that may require significant time or space to fully explain to someone who is not familiar with the associated background literature. However, for practical reasons, questionnaires contain short definitions. Consequently, shortening a definition simply for the sake of reducing the reading time was not a goal for the attached items. If anything, some people could urge that definitions could be *longer*. However, reducing the amount of confusing, extraneous, or unnecessarily complex text was a goal.
- The job incumbent may not always be the most appropriate source of the ratings for abilities and styles, and perhaps skills as well. An appropriate rater would already have some knowledge of these domains in terms of what these constructs mean and how they are typically assessed (e.g., Flexibility of Closure, Perceptual Speed). Even people who have researched these characteristics of individual differences still have disagreements over what is the optimal characterization for some of them. Some job incumbents may find making judgments based on the current definitions and scale anchors to be a difficult task.
- Another issue is whether the job incumbents should describe their own personal job situation or attempt to describe what the job/occupation is like in general (i.e., for job holders in general). It is probably an easier judgment task to describe your own job situation. However, the O*NET model wants descriptions of the general characteristic nature of the occupations, and the directions for the current versions of the instruments ask the rater to use a general frame-of-reference. Asking for descriptions of the respondents' own job would be a major change to the directions for the instruments, with unknown effects. Such effects could be positive, negative, or neutral. If the sample of incumbents is reasonable, the frame-of-reference effects shouldn't be that great, but that is not known at this point. The goal is to minimize within-occupation variance and to maximize meaningful cross-occupation variance.

- The redundancy across domain questionnaires was another general issue. For example, Creativity or Idea Generation can be found in Abilities, Work Styles, Skills, and Generalized Work Activities. Problem solving can be found in Work Styles, Skills, and Generalized Work Activities. At some point the O*NET Model should get clearer about the distinctions among domains, and proceed to deal with these redundancies.

Summary of Substantive Questionnaire Revisions

Table 2 summarizes the revisions that were proposed and adopted for the seven principal taxonomies in the O*NET Content Model. Two kinds of revisions are reflected in the table: 1) reductions in the number of items and 2) reductions in the number of response scales per item. Minor wording changes were made to some of the specific variable definitions, but none of these represented substantive alterations and thus, these kinds of minor edits are not reflected in the table. In all cases, wording changes were agreed upon by the O*NET Consortium panel and DOL representatives. In ten instances (out of over 250 items), there were minor changes to the title of the variables. These changes in the descriptor title were always for the purpose of making the title more correspondent with the definition.

Table 2. O*NET Questionnaire Item (Descriptor Variable) Revisions

Measurement Questionnaire	# of Items in Revised Version	# of Items in Original Version	Response Scales
Work Values	0	21	Dropped from current O*NET questionnaire
Abilities	52	52	No changes
Knowledge	33	33	Keep Level and Importance, Drop Job Specialty Requirements
Education & Training	5	15	Drop Instructional Program Required Drop Level of Education Required in Specific Subject Areas Drop Licensure, Certification, and Registration
Work Styles	16	17	Keep Importance Scale, Drop Level Scale
Skills	35	46	Keep Level and Importance, Drop Job Entry Requirements
Generalized Work Activities	41	42	Keep Level and Importance, Drop Frequency
Work Context	<u>57</u>	<u>97</u>	Adopt consistent use of 5-point scales
TOTAL ITEMS	239	323	

There are several points to be noted about the table.

- The Occupational Values questionnaire is not being administered to job incumbents in the pretest. This questionnaire is taken directly from the Minnesota Theory of Work Adjustment (Dawis and Lofquist, 1984). When used to assess the incentive or reinforcing properties of jobs or occupations, the instrument is always completed by supervisors and yields the Occupational Reinforcer Pattern (ORP) for the occupation. When used by job incumbents, it is viewed as an indicator of individual job satisfaction (Dawis and Lofquist, 1984).
- The reduction of the number of Skill descriptors from 46 to 35 is solely a function of making certain item combinations. Job Entry Requirements were dropped from the Skills Questionnaire because job incumbents tend to be poor sources of information for this.
- The checklist of “Instructional Programs Required” and the Level of Education Required in “15 Educational Subject Areas” checklist do not appear here, as is the case with “Job Specialty Requirements” in the Knowledges Questionnaire. These data for occupations can be obtained from other sources (i.e., data collected by other organizations).
- The greatest number of item deletions were made on the Work Context Questionnaire. These will be detailed in the Work Context section. For six items, two of the three response scales were dropped. Consequently, these six are counted as six, and not 18 for purposes of representing the total number of items. The total item count in the table may not correspond precisely to other item counts. For example, in the original version, if the multiple responses per item are included the total number of judgments is 97.
- The Level scale was eliminated from the Work Styles Questionnaire due to previous findings that it was highly redundant with the Importance scale.
- The Frequency scale was dropped from the Generalized Work Activities Questionnaire because it has been found to be highly correlated with Importance and also redundant with Frequency scale ratings for tasks.

Appendix A contains documentation of the specific changes for each questionnaire. First, the original version of the item/descriptor is presented, then the modified version is shown. For some items, there were no changes, or only minor wording changes. For a smaller number of items, more substantial changes were made. For these items, the reasons/rationale for the change are given.

The outcomes of the item edits are described below.

- Appendix A presents all the variable (item) titles and definitions for Abilities, Generalized Work Activities, Knowledge, Education and Training, Work Styles, Skills, and Work Context. That is, no items are omitted. The attached versions represent the results of synthesizing the suggestions of Instrument Revision Working Group members for semantic/stylistic clarification and simplification. Some items changed a fair amount, some changed a little bit and some changed not at all. The substantive meanings were not changed, only clarified.
- The domain frame-of-reference (as in the ability to, knowledge of, etc.) was retained as part of each variable definition even though it is redundant across the items in a domain. We believe that the respondents need all the help they can get in trying to keep clear what kind of characteristic is being rated.

Chapter 5. Pilot Testing

As drafts of the revised instruments were produced, they were tested to evaluate the results of modifications. Testing also enabled the revision of estimates for the time needed to complete the various instruments. Such information was crucial to the design of an effective survey and for obtaining approval from the Office of Management and Budget (OMB) for conducting subsequent data collection.

Procedures

A small test of the questionnaires was conducted in August, 1998, to obtain completion times to individual questionnaires. The average times are based on from 2 to 7 people answering the questionnaires. Because some questionnaires were not ready until the timing tests were already under way, completion times are based on only two or three raters. Participants in the timing exercise were 11 RTI employees from a variety of trade jobs (carpenter, HVAC technician, plumber, electrician) as well as clerks and supervisors from RTI's data preparation department. Not all respondents had task lists (a suitable match could not be found for the data preparation jobs), but all completed at least 2 instruments.

Results

Table 3 contains the results of the completion time testing. Included in Table 3 are the estimated times for completion of the prototype instruments. These times were obtained from the Clearance Package submitted to the OMB for previous testing of the prototype instruments. The scope of the completion time testing did not allow rigorous testing of the instruments under field conditions. However, the shorter completion times for the revised instruments compared to the prototype instruments indicate that the revisions were successful at reducing the burden on respondents. However, only more thorough testing of the revised instruments can demonstrate that they are also effective at measuring the information necessary to meet the needs of the content model and O*NET more generally.

Table 3. Completion Times for O*NET Questionnaires (Minutes:Seconds)

QUESTIONNAIRE	Completion time for Prototype Instrument ¹	Number of Raters	Completion Times from Tests of Revised Instruments		
			Mean	Min	Max
Work Style	8:48	4	3:67	2:00	6:00
Work Context	23:00	3	14:33	13:00	16:00
Generalized Work Activities	20:00	4	13:50	10:00	15:00
Abilities	24:00	5	16:80	10:00	25:00
Skills	27:30	2	17:00	15:00	19:00
Knowledge	14:00	2	12:00	10:00	14:00

¹ Obtained from the Clearance Package submitted to the OMB for previous testing of the prototype instruments.

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Appendix A

Changes in Titles and Wording of Definitions in Abilities, Generalized Work Activities, Knowledge, Education and Training, Work Styles, Skills, and Work Context Questionnaires

Abilities

Original

Revision

- | | |
|--|--|
| 1. Oral Comprehension
The ability to listen to and understand information and ideas presented through spoken words and sentences. | 1. No change. |
| 2. Written Comprehension
The ability to read and understand information and ideas presented in writing. | 2. No change. |
| 3. Oral Expression
The ability to communicate information and ideas in speaking so others will understand. | 3. No change. |
| 4. Written Expression
The ability to communicate information and ideas in writing so others will understand. | 4. No change. |
| 5. Fluency of Ideas
The ability to come up with a number of ideas about a given topic. It concerns the number of ideas produced and <u>not</u> the quality, correctness, or creativity of the ideas. | 5. Fluency of Ideas (Minimal)
The ability to come up with a number of ideas about a topic (the <i>number</i> of ideas is important, <u>not</u> their quality, correctness, or creativity). |
| 6. Originality
The ability to come up with unusual or clever ideas about a given topic or situation, or to develop creative ways to solve a problem. | 6. No change. |
| 7. Problem Sensitivity
The ability to tell when something is wrong or is likely to go wrong. It does <u>not</u> involve solving the problem, only recognizing that there is a problem. | 7. No change. |
| 8. Deductive Reasoning
The ability to apply general rules to specific problems to come up with logical answers. It involves deciding if an answer makes sense. | 8. Deductive Reasoning (Minimal)
The ability to apply general rules to specific problems to produce answers that make sense. |

Abilities

Original

9. Inductive Reasoning

The ability to combine separate pieces of information, or specific answers to problems, to form general rules or conclusions. It includes coming up with a logical explanation for why a series of seemingly unrelated events occur together.

10. Information Ordering

The ability to correctly follow a given rule or set of rules in order to arrange things or actions in a certain order. The things or actions can include numbers, letters, words, pictures, procedures, sentences, and mathematical or logical operations.

11. Category Flexibility

The ability to produce many rules so that each rule tells how to group (or combine) a set of things in a different way.

12. Mathematical Reasoning

The ability to understand and organize a problem and then to select a mathematical method or formula to solve the problem.

13. Number Facility

The ability to add, subtract, multiply, or divide quickly and correctly.

14. Memorization

The ability to remember information such as words, numbers, pictures, and procedures.

15. Speed of Closure

The ability to quickly make sense of information that seems to be without meaning or organization. It involves quickly combining and organizing different pieces of information into a meaningful pattern.

Revision

9. Inductive Reasoning (Minimal)

The ability to combine pieces of information to form general rules or conclusions (includes finding a relationship among seemingly unrelated events).

10. Information Ordering (Minimal)

The ability to arrange things or actions in a certain order or pattern according to a specific rule or set of rules (e.g., patterns of numbers, letters, words, pictures, mathematical operations).

11. Category Flexibility (Minimal)

The ability to generate or use different sets of rules for combining or grouping things in different ways.

12. Mathematical Reasoning (Minimal)

The ability to choose the right mathematical methods or formulas to solve a problem.

13. No change.

14. No change.

15. Speed of Closure (Minimal)

The ability to quickly make sense of, combine, and organize information into meaningful patterns.

Abilities

Original

16. Flexibility of Closure

The ability to identify or detect a known pattern (a figure, object, word, or sound) that is hidden in other distracting material.

17. Perceptual Speed

The ability to quickly and accurately compare letters, numbers, objects, pictures, or patterns. The things to be compared may be presented at the same time or one after the other. This ability also includes comparing a presented object with a remembered object.

18. Spatial Orientation

The ability to know one's location in relation to the environment, or to know where other objects are in relation to one's self.

19. Visualization

The ability to imagine how something will look after it is moved around or when its parts are moved or rearranged.

20. Selective Attention

The ability to concentrate and not be distracted while performing a task over a period of time.

21. Time Sharing

The ability to efficiently shift back and forth between two or more activities or sources of information (such as speech, sounds, touch, or other sources).

22. Arm-Hand Steadiness

The ability to keep the hand and arm steady while making an arm movement or while holding the arm and hand in one position.

Revision

16. No change.

17. Perceptual Speed (Minimal)

The ability to quickly and accurately compare similarities and differences among sets of letters, numbers, objects, pictures, or patterns. The things to be compared may be presented at the same time or one after the other. This ability also includes comparing a presented object with a remembered object.

18. Spatial Orientation (Minimal)

The ability to know your location in relation to the environment or to know where other objects are in relation to you.

19. No change.

20. Selective Attention (Minimal)

The ability to concentrate on a task over a period of time without being distracted.

21. Time Sharing (Minimal)

The ability to shift back and forth between two or more activities or sources of information (such as speech, sounds, touch, or other sources).

22. Arm-Hand Steadiness (Minimal)

The ability to keep your hand and arm steady while moving your arm or while holding your arm and hand in one position.

Abilities

Original

23. Manual Dexterity

The ability to quickly make coordinated movements of one hand, a hand together with the arm, or two hands to grasp, manipulate, or assemble objects.

24. Finger Dexterity

The ability to make precisely coordinated movements of the fingers of one or both hands to grasp, manipulate, or assemble very small objects.

25. Control Precision

The ability to quickly and repeatedly make precise adjustments in moving the controls of a machine or vehicle to exact positions.

26. Multilimb Coordination

The ability to coordinate movements of two or more limbs together (for example, two arms, two legs, or one leg and one arm) while sitting, standing, or lying down. It does not involve performing the activities while the body is in motion.

27. Response Orientation

The ability to choose quickly and correctly between *two or more movements* in response to *two or more different signals* (lights, sounds, pictures, etc.). It includes the *speed* with which the correct response is *started* with the hand, foot, or other body parts.

28. Rate Control

The ability to time the adjustments of a movement or equipment control in anticipation of changes in the speed and/or direction of a continuously moving object or scene.

Revision

23. Manual Dexterity (Minimal)

The ability to quickly move your hand, your hand together with your arm, or your two hands to grasp, manipulate, or assemble objects.

24. No change.

25. Control Precision (Minimal)

The ability to quickly and repeatedly adjust the controls of a machine or a vehicle to exact positions.

26. Multilimb Coordination (Minimal)

The ability to coordinate two or more limbs (for example, two arms, two legs, or one leg and one arm) while sitting, standing, or lying down. It does not involve performing the activities while the whole body is in motion.

27. Response Orientation (Minimal)

The ability to choose quickly between *two or more movements* in response to *two or more different signals* (lights, sounds, pictures). It includes the speed with which the correct response is *started* with the hand, foot, or other body part.

28. Rate Control (Minimal)

The ability to time your movements or the movement of a piece of equipment in anticipation of changes in the speed and/or direction of a moving object or scene.

Abilities

Original

29. Reaction Time

The ability to quickly respond (with the hand, finger, or foot) to one signal (sound, light, picture, etc.) when it appears.

30. Wrist-Finger Speed

The ability to make *fast, simple*, repeated movements of the *fingers, hands, and wrists*.

31. Speed of Limb Movement

The ability to *quickly* move the arms or legs.

32. Static Strength

The ability to exert maximum muscle force to lift, push, pull, or carry objects.

33. Explosive Strength

The ability to use short bursts of muscle force to propel oneself (as in jumping or sprinting) or to throw an object.

34. Dynamic Strength

The ability to exert muscle force repeatedly or continuously over time. This involves muscular endurance and resistance to muscle fatigue.

35. Trunk Strength

The ability to use one's abdominal and lower back muscles to support part of the body repeatedly or continuously over time without "giving out" or fatiguing.

36. Stamina

The ability to exert one's self physically over long periods of time without getting winded or out of breath.

Revision

29. Reaction Time (Minimal)

The ability to quickly respond (with the hand, finger, or foot) to a signal (sound, light, picture) when it appears.

30. No change.

31. Speed of Limb Movement (Minimal)

The ability to *quickly* move the arms and legs.

32. No change.

33. No change.

34. No change.

35. Trunk Strength (Minimal)

The ability to use your abdominal and lower back muscles to support part of the body repeatedly or continuously over time without "giving out" or fatiguing.

36. Stamina (Minimal)

The ability to exert yourself physically over long periods of time without getting winded or out of breath.

Abilities

Original

- 37. Extent Flexibility**
The ability to bend, stretch, twist, or reach out with the body, arms, and/or legs.
- 38. Dynamic Flexibility**
The ability to quickly and repeatedly bend, stretch, twist, or reach out with the body, arms, and/or legs.
- 39. Gross Body Coordination**
The ability to coordinate the *movement of the arms, legs, and torso together* in activities where the whole body is in motion.
- 40. Gross Body Equilibrium**
The ability to keep or regain one's body balance or stay upright when in an unstable position.
- 41. Near Vision**
The ability to see details of objects at a close range (within a few feet of the observer).
- 42. Far Vision**
The ability to see details at a distance.
- 43. Visual Color Discrimination**
The ability to match or detect differences between colors, including shades of color and brightness.
- 44. Night Vision**
The ability to see under low-light conditions.
- 45. Peripheral Vision**
The ability to see objects or movement of objects to one's side when the eyes are focused forward.

Revision

- 37. Extent Flexibility (Minimal)**
The ability to bend, stretch, twist, or reach with your body, arms, and/or legs.
- 38. Dynamic Flexibility (Minimal)**
The ability to quickly and repeatedly bend, stretch, twist, or reach out with your body, arms, and/or legs.
- 39. Gross Body Coordination (Minimal)**
The ability to coordinate the *movement of your arms, legs, and torso together* when the whole body is in motion.
- 40. Gross Body Equilibrium (Minimal)**
The ability to keep or regain your body balance or stay upright when in an unstable position.
- 41. Near Vision (Minimal)**
The ability to see details at close range (within a few feet of the observer).
- 42.** No change.
- 43.** No change.
- 44.** No change.
- 45. Peripheral Vision (Minimal)**
The ability to see objects or movement of objects to one's side when the eyes are looking ahead.

Abilities

Original

46. Depth Perception

The ability to judge which of several objects is closer or farther away from the observer, or to judge the distance between an object and the observer.

47. Glare Sensitivity

The ability to see objects in the presence of a glare or bright lighting.

48. Hearing Sensitivity

The ability to detect or tell the difference between sounds that vary over broad ranges of pitch and loudness.

49. Auditory Attention

The ability to focus on a single source of auditory (hearing) information in the presence of other distracting sounds.

50. Sound Localization

The ability to tell the direction from which a sound originated.

51. Speech Recognition

The ability to identify and understand the speech of another person.

52. Speech Clarity

The ability to speak clearly so that it is understandable to a listener.

Revision

46. Depth Perception (Minimal)

The ability to judge which of several objects is closer or farther away from you, or to judge the distance between you and an object.

47. No change.

48. Hearing Sensitivity (Minimal)

The ability to detect or tell the differences between sounds that vary in pitch and loudness.

49. Auditory Attention (Minimal)

The ability to focus on a single source of sound in the presence of other distracting sounds.

50. No change.

51. No change.

52. Speech Clarity (Minimal)

The ability to speak clearly so others can understand you.

Generalized Work Activities

Original

1. **Getting Information Needed to Do the Job**
Observing, receiving, and otherwise obtaining information from all relevant sources.
2. **Identifying Objects, Actions, and Events**
Identifying information received by making estimates or categorizations, recognizing differences or similarities, or sensing changes in circumstances or events.
3. **Monitoring Processes, Materials, or Surroundings**
Monitoring and reviewing information from materials, events, or the environment, often to detect problems or to find out when things are finished.
4. **Inspecting Equipment, Structures, or Materials**
Inspecting or diagnosing equipment, structures, or materials to identify the causes of errors or other problems or defects.
5. **Estimating the Characteristics of Materials, Products, Events, or Information**
Estimating sizes, distances, and quantities, or determining time, costs, resources, or materials needed to perform a work activity.
6. **Judging the Qualities of Objects, Services, or Persons**
Making judgments about or assessing the value, importance, or quality of things or people.

Revision

1. **Getting Information** (Minimal)
Observing, receiving, and otherwise obtaining information from all relevant sources.
2. **Identifying Objects, Actions, and Events** (Minimal)
Identifying information by categorizing, estimating, recognizing differences or similarities, and detecting changes in circumstances or events.
3. **Monitoring Processes, Materials, or Surroundings** (Minimal)
Monitoring and reviewing information from materials, events, or the environment, to detect or assess problems.
4. **Inspecting Equipment, Structures, or Materials** (Minimal)
Inspecting equipment, structures, or materials to identify the cause of errors or other problems or defects.
5. **Estimating the Quantifiable Characteristics of Products, Events, or Information** (Minimal)
Estimating sizes, distances, and quantities; or determining time, costs, resources, or materials needed to perform a work activity.
6. **Judging the Qualities of Objects, Services, or People** (Minimal)
Assessing the value, importance, or quality of things or people.

Generalized Work Activities

Original

7. Evaluating Information for Compliance to Standards

Evaluating information against a set of standards and verifying that it is correct.

8. Processing Information

Compiling, coding, categorizing, calculating, tabulating, auditing, verifying, or processing information or data.

9. Analyzing Data or Information

Identifying underlying principles, reasons, or facts by breaking down information or data into separate parts.

10. Making Decisions and Solving Problems

Combining, evaluating, and reasoning with information and data to make decisions and solve problems. These processes involve making decisions about the relative importance of information and choosing the best solution.

11. Thinking Creatively

Originating, inventing, designing, or creating new applications, ideas, relationships, systems, or products, including artistic contributions.

12. Updating and Using Job-Relevant Knowledge

Keeping up-to-date technically and knowing one's own jobs' and related jobs' functions.

Revision

7. Evaluating Information to Determine Compliance with Standards (Moderate)

Using relevant information and individual judgment to determine whether events or processes comply with laws, regulations, or standards.

8. Processing Information (Minimal)

Compiling, coding, categorizing, calculating, tabulating, auditing, or verifying information or data.

9. Analyzing Data or Information (Minimal)

Identifying the underlying principles, reasons, or facts of information by breaking down information or data into separate parts.

10. Making Decisions and Solving Problems (Minimal)

Analyzing information and evaluating results to choose the best solution and solve problems.

11. Thinking Creatively (Minimal)

Developing, designing, or creating new applications, ideas, relationships, systems, or products, including artistic contributions.

12. Updating and Using Relevant Knowledge (Moderate)

Keeping up-to-date technically and applying new knowledge to your job.

Generalized Work Activities

Original

13. Developing Objectives and Strategies

Establishing long-range objectives and specifying the strategies and actions to achieve these objectives.

14. Scheduling Work and Activities

Scheduling events, programs, activities, as well as the work of others.

15. Organizing, Planning, and Prioritizing Work

Developing plans to accomplish work, and prioritize and organizing one's own work.

16. Performing General Physical Activities

Performing physical activities that require moving one's whole body, such as in climbing, lifting, balancing, walking, stooping, where the activities often also require considerable use of the arms and legs, such as in the physical handling of materials.

17. Handling and Moving Objects

Using one's own hands and arms in handling, installing, positioning, and moving materials, or in manipulating things, including the use of keyboards.

18. Controlling Machines and Processes

Using either control mechanisms or direct physical activity to operate machines or processes (not including computers or vehicles).

Revision

13. Developing Objectives and Strategies

(Minimal)

Establishing long-range objectives and specifying the strategies and actions to achieve them.

14. Scheduling Work and Activities

(Minimal)

Scheduling events, programs, and activities, as well as the work of others.

15. Organizing, Planning, and Prioritizing Work (Minimal)

Developing specific goals and plans to prioritize, organize, and accomplish your work.

16. Performing General Physical Activities

(Minimal)

Performing physical activities that require considerable use of your arms and legs and moving your whole body, such as climbing, lifting, balancing, walking, stooping, and handling of materials.

17. Handling and Moving Objects (Minimal)

Using hands and arms in handling, installing, positioning, and moving materials, and manipulating things.

18. No change.

Generalized Work Activities

Original

19. Interacting with Computers

Controlling computer functions by using programs, setting up functions, writing software, or otherwise communicating with computer systems.

20. Operating Vehicles, Mechanized Devices, or Equipment

Running, maneuvering, navigating, or driving vehicles or mechanized equipment, such as forklifts, passenger vehicles, aircraft, or water craft.

21. Drafting, Laying-Out, and Specifying Technical Devices, Parts, or Equipment

Providing documentation, detailed instructions, drawings, or specifications to inform others about how devices, parts, equipment, or structures are to be fabricated, constructed, assembled, modified, maintained, or used.

22. Implementing Ideas, Programs, Systems, or Products

Carrying out work activities using one's own ideas or directions and instructions from others to install, prepare, deliver, construct, or complete products or systems.

23. Repairing and Maintaining Mechanical Equipment

Fixing, servicing, aligning, setting up, adjusting, and testing machines, devices, moving parts, and equipment that operate primarily on the basis of mechanical (not electronic) principles.

Revision

19. Working with Computers (Minimal)

Using computers and computer systems (including hardware and software) to program, write software, set up functions, enter data, or process information.

20. No change.

21. Drafting, Laying Out, and Specifying Technical Devices, Parts, and Equipment (Minimal)

Providing documentation, detailed instructions, drawings, or specifications to tell others about how devices, parts, equipment, or structures are to be fabricated, constructed, assembled, modified, maintained, or used.

Delete.

22. Repairing and Maintaining Mechanical Equipment (Minimal)

Servicing, repairing, adjusting, and testing machines, devices, moving parts, and equipment that operate primarily on the basis of mechanical (not electronic) principles.

Generalized Work Activities

Original

24. Repairing and Maintaining Electronic Equipment

Fixing, servicing, adjusting, regulating, calibrating, fine-tuning, or testing machines, devices, and equipment that operate primarily on the basis of electrical or electronic (not mechanical) principles.

25. Documenting/Recording Information

Entering, transcribing, recording, storing, or maintaining information in either written form or by electronic/magnetic recording.

26. Interpreting the Meaning of Information for Others

Translating or explaining what information means and how it can be understood or used to support responses or feedback to others.

27. Communicating with Supervisors, Peers, or Subordinates

Providing information to supervisors, fellow workers, and subordinates. This information can be exchanged face-to-face, in writing, or via telephone/electronic transfer.

28. Communicating with Persons Outside the Organization

Communicating with persons outside the organization, representing the organization to customers, the public, government, and other external sources. This information can be exchanged face-to-face, in writing, or via telephone/electronic transfer.

Revision

23. Repairing and Maintaining Electronic Equipment (Minimal)

Servicing, repairing, calibrating, regulating, fine-tuning, or testing machines, devices, and equipment that operate primarily on the basis of electrical or electronic (not mechanical) principles.

24. Documenting/Recording Information (Minimal)

Entering, transcribing, recording, storing, or maintaining information in written or electronic/magnetic form.

25. Interpreting the Meaning of Information for Others (Minimal)

Translating or explaining what information means and how it can be used.

26. Communicating with Supervisors, Peers, or Subordinates (Minimal)

Providing information to supervisors, co-workers, and subordinates by telephone, in written form, e-mail, or in person.

27. Communicating with People Outside the Organization (Minimal)

Communicating with people outside the organization, representing the organization to customers, the public, government, and other external sources. This information can be exchanged in person, in writing, or by telephone or e-mail.

Generalized Work Activities

Original

- 29. Establishing and Maintaining Interpersonal Relationships**
Developing constructive and cooperative working relationships with others.
- 30. Assisting and Caring for Others**
Providing assistance or personal care to others.
- 31. Selling or Influencing Others**
Convincing others to buy merchandise/goods, or otherwise changing their minds or actions.
- 32. Resolving Conflicts and Negotiating with Others**
Handling complaints, arbitrating disputes, and resolving grievances, or otherwise negotiating with others.
- 33. Performing for or Working Directly with the Public**
Performing for people or dealing directly with the public, including serving persons in restaurants and stores, and receiving clients or guests.
- 34. Coordinating the Work and Activities of Others**
Coordinating members of a work group to accomplish tasks.
- 35. Developing and Building Teams**
Encouraging and building mutual trust, respect, and cooperation among team members.

Revision

- 28. Establishing and Maintaining Interpersonal Relationships (Minimal)**
Developing constructive and cooperative working relationships with others, and maintaining them over time.
- 29. Assisting and Caring for Others (Moderate)**
Providing personal assistance, medical attention, emotional support, or other personal care to others such as coworkers, customers, or patients.
- 30. Selling or Influencing Others (Minimal)**
Convincing others to buy merchandise/goods or to otherwise change their minds or actions.
- 31. Resolving Conflicts and Negotiating with Others (Minimal)**
Handling complaints, settling disputes, and resolving grievances and conflicts, or otherwise negotiating with others.
- 32. Performing for or Working Directly with the Public (Minimal)**
Performing for people or dealing directly with the public. This includes serving customers in restaurants and stores, and receiving clients or guests.
- 33. Coordinating the Work and Activities of Others (Minimal)**
Getting members of a group to work together to accomplish tasks.
- 34.** No change.

Generalized Work Activities

Original

36. Teaching Others

Identifying educational needs, developing formal training programs or classes, and teaching or instructing others.

37. Guiding, Directing, and Motivating Subordinates

Providing guidance and direction to subordinates, including setting performance standards and monitoring subordinates.

38. Coaching and Developing Others

Identifying developmental needs of others and coaching or otherwise helping others to improve their knowledge or skills.

39. Providing Consultation and Advice to Others

Providing consultation and expert advice to management or other groups on technical, systems-related, or process-related topics.

40. Performing Administrative Activities

Approving requests, handling paperwork, and performing day-to-day administrative tasks.

41. Staffing Organizational Units

Recruiting, interviewing, selecting, hiring, and promoting persons for an organization.

42. Monitoring and Controlling Resources

Monitoring and controlling resources and overseeing the spending of money.

Revision

35. Training and Teaching Others

(Minimal)

Identifying the educational needs of others, developing formal educational or training programs or classes, and teaching or instructing others.

36. Guiding, Directing, and Motivating Subordinates (Minimal)

Providing guidance and direction to subordinates, including setting performance standards and monitoring performance.

37. Coaching and Developing Others

(Minimal)

Identifying the developmental needs of others and coaching, mentoring, or otherwise helping others to improve their knowledge or skills.

38. Providing Consultation and Advice to Others (Minimal)

Providing guidance and expert advice to management or other groups on technical, systems-, or process-related topics.

39. Performing Administrative Activities

(Minimal)

Performing day-to-day administrative tasks such as maintaining information files and processing paperwork.

40. Staffing Organizational Units (Minimal)

Recruiting, interviewing, selecting, hiring, and promoting employees in an organization.

41. No change.

Knowledge

Original

- 1. Administration and Management**
Knowledge of principles and processes involved in business and organizational planning, coordination, and execution. This includes strategic planning, resource allocation, manpower modeling, leadership techniques, and production methods.
- 2. Clerical**
Knowledge of administrative and clerical procedures and systems such as word processing systems, filing and records management systems, stenography and transcription, forms design principles, and other office procedures and terminology.
- 3. Economics and Accounting**
Knowledge of economic and accounting principles and practices, the financial markets, banking and the analysis and reporting of financial data.
- 4. Sales and Marketing**
Knowledge of principles and methods involved in showing, promoting, and selling products or services. This includes marketing strategies and tactics, product demonstration and sales techniques, and sales control systems.
- 5. Customer and Personal Service**
Knowledge of principles and processes for providing customer and personal services including needs assessment techniques, quality service standards, alternative delivery systems, and customer satisfaction evaluation techniques.

Revision

- 1. Administration and Management**
(Minimal)
Knowledge of business and management principles involved in strategic planning, resource allocation, human resources modeling, leadership technique, production methods, and coordination of people and resources.
- 2. Clerical** (Minimal)
Knowledge of administrative and clerical procedures and systems such as word processing, managing files and records, stenography and transcription, designing forms, and other office procedures and terminology.
- 3.** No change.
- 4. Sales and Marketing** (Minimal)
Knowledge of principles and methods for showing, promoting, and selling products or services. This includes marketing strategy and tactics, product demonstration, sales techniques, and sales control systems.
- 5. Customer and Personal Service**
(Minimal)
Knowledge of principles and processes for providing customer and personal services. This includes customer needs assessment, meeting quality standards for services, and evaluation of customer satisfaction.

Knowledge

Original

- 6. Personnel and Human Resources**
Knowledge of policies and practices involved in personnel/human resource functions. This includes recruitment, selection, training, and promotion regulations and procedures; compensation and benefits packages; labor relations and negotiation strategies; and personnel information systems.
- 7. Production and Processing**
Knowledge of inputs, outputs, raw materials, waste, quality control, costs, and techniques for maximizing the manufacture and distribution of goods.
- 8. Food Production**
Knowledge of techniques and equipment for planting, growing, and harvesting of food for consumption including crop rotation methods, animal husbandry, and food storage/handling techniques.
- 9. Computers and Electronics**
Knowledge of electric circuit boards, processors, chips, and computer hardware and software, including applications and programming.
- 10. Engineering and Technology**
Knowledge of equipment, tools, mechanical devices, and their uses to produce motion, light, power, technology, and other applications.

Revision

- 6. Personnel and Human Resources (Minimal)**
Knowledge of principles and procedures for personnel recruitment, selection, training, compensation and benefits, labor relations and negotiation, and personnel information systems.
- 7. Production and Processing (Minimal)**
Knowledge of raw materials, production processes, quality control, costs, and other techniques for maximizing the effective manufacture and distribution of goods.
- 8. Food Production (Minimal)**
Knowledge of techniques and equipment for planting, growing, and harvesting food products (both plant and animal) for consumption, including storage/handling techniques.
- 9. Computers and Electronics (Minimal)**
Knowledge of circuit boards, processors, chips, electronic equipment, and computer hardware and software, including applications and programming.
- 10. Engineering and Technology (Moderate)**
Knowledge of the practical application of engineering science and technology. This includes applying principles, techniques, procedures, and equipment to the design and production of various goods and services.

Knowledge

Original

11. Design

Knowledge of design techniques, principles, tools and instruments involved in the production and use of precision technical plans, blueprints, drawings, and models.

12. Building and Construction

Knowledge of materials, methods, and the appropriate tools to construct objects, structures, and buildings.

13. Mechanical

Knowledge of machines and tools, including their designs, uses, benefits, repair, and maintenance.

14. Mathematics

Knowledge of numbers, their operations, and interrelationships including arithmetic, algebra, geometry, calculus, statistics, and their applications.

15. Physics

Knowledge and prediction of physical principles, laws, and applications including air, water, material dynamics, light, atomic principles, heat, electric theory, earth formations, and meteorological and related natural phenomena.

16. Chemistry

Knowledge of the composition, structure, and properties of substances and of the chemical processes and transformations that they undergo. This includes uses of chemicals and their interactions, danger signs, production techniques, and disposal methods.

Revision

11. Design (Minimal)

Knowledge of design techniques, tools, and principles involved in production of precision technical plans, blueprints, drawings, and models.

12. Building and Construction (Minimal)

Knowledge of materials, methods, and the tools involved in the construction or repair of houses, buildings, or other structures such as highways and roads.

13. Mechanical (Minimal)

Knowledge of machines and tools, including their designs, uses, repair, and maintenance.

14. Mathematics (Minimal)

Knowledge of arithmetic, algebra, geometry, calculus, statistics, and their applications.

15. Physics (Minimal)

Knowledge and prediction of physical principles, laws, their interrelationships, and applications to understanding fluid, material, and atmospheric dynamics, and mechanical, electrical, atomic, and sub-atomic structures and processes.

16. Chemistry (Minimal)

Knowledge of the chemical composition, structure, and properties of substances and of the chemical processes and transformations that they undergo. This includes uses of chemicals and their interactions, danger signs, production techniques, and disposal methods.

Knowledge

Original

17. Biology

Knowledge of plant and animal living tissue, cells, organisms, and entities, including their functions, interdependencies, and interactions with each other and the environment.

18. Psychology

Knowledge of human behavior and performance, mental processes, psychological research methods, and the assessment and treatment of behavioral and affective disorders.

19. Sociology and Anthropology

Knowledge of group behavior and dynamics, societal trends and influences, cultures, their history, migrations, ethnicity, and origins.

20. Geography

Knowledge of various methods for describing the location and distribution of land, sea, and air masses including their physical locations, relationships, and characteristics.

21. Medicine and Dentistry

Knowledge of the information and techniques needed to diagnose and treat injuries, diseases, and deformities. This includes symptoms, treatment alternatives, drug properties and interactions, and preventive health-care measures.

Revision

17. Biology (Minimal)

Knowledge of plant and animal organisms, their tissues, cells, functions, interdependencies, and interactions with each other and the environment.

18. Psychology (Minimal)

Knowledge of human behavior and performance; individual differences in ability, personality, and interests; learning and motivation; psychological research methods; and the assessment and treatment of behavioral and affective disorders.

19. Sociology and Anthropology (Minimal)

Knowledge of group behavior and dynamics, societal trends and influences, human migrations, ethnicity, cultures and their history and origins.

20. Geography (Minimal)

Knowledge of principles and methods for describing the features of land, sea, and air masses, including their physical characteristics, locations, interrelationships, and distribution of plant, animal, and human life.

21. Medicine and Dentistry (Minimal)

Knowledge of the information and techniques needed to diagnose and treat human injuries, diseases, and deformities. This includes symptoms, treatment alternatives, drug properties and interactions, and preventive health-care measures.

Knowledge

Original

22. Therapy and Counseling

Knowledge of information and techniques needed to rehabilitate physical and mental ailments, and to provide career guidance including alternative treatments, rehabilitation equipment and its proper use, and methods to evaluate treatment effects.

23. Education and Training

Knowledge of instructional methods and training techniques including curriculum design principles, learning theory, group and individual teaching techniques, design of individual development plans, and test design principles.

24. English Language

Knowledge of the structure and content of the English language including the meaning and spelling of words, rules of composition, and grammar.

25. Foreign Language

Knowledge of the structure and content of a foreign (non-English) language including the meaning and spelling of words, rules of composition and grammar, and pronunciation.

26. Fine Arts

Knowledge of theory and techniques required to produce, compose, and perform works of music, dance, visual arts, drama, and sculpture.

27. History and Archeology

Knowledge of past historical events and their causes, indicators, and impact on particular civilizations and cultures.

Revision

22. Therapy and Counseling (Minimal)

Knowledge of principles, methods, and procedures for diagnosis, treatment, and rehabilitation of physical and mental dysfunctions, and for career counseling and guidance.

23. Education and Training (Minimal)

Knowledge of principles and methods for curriculum and training design, teaching and instruction for individuals and groups, and the measurement of training effects.

24. No change.

25. No change.

26. Fine Arts (Minimal)

Knowledge of the theory and techniques required to compose, produce, and perform works of music, dance, visual arts, drama, and sculpture.

27. History and Archeology (Minimal)

Knowledge of historical events and their causes, indicators, and effects on civilizations and cultures.

Knowledge

Original

28. Philosophy and Theology

Knowledge of different philosophical systems and religions, including their basic principles, values, ethics, ways of thinking, customs, and practices, and their impact on human culture.

29. Public Safety and Security

Knowledge of weaponry, public safety, and security operations, rules, regulations, precautions, prevention, and the protection of people, data, and property.

30. Law, Government and Jurisprudence

Knowledge of laws, legal codes, court procedures, precedents, government regulations, executive orders, agency rules, and the democratic political process.

31. Telecommunications

Knowledge of transmission, broadcasting, switching, control, and operation of telecommunications systems.

32. Communications and Media

Knowledge of media production, communication, and dissemination techniques and methods including alternative ways to inform and entertain via written, oral, and visual media.

33. Transportation

Knowledge of principles and methods for moving people or goods by air, rail, sea, or road, including their relative costs, advantages, and limitations.

Revision

28. Philosophy and Theology (Minimal)

Knowledge of different philosophical systems and religions. This includes their basic principles, values, ethics, ways of thinking, customs, practices, and their impact on human culture.

29. Public Safety and Security (Minimal)

Knowledge of relevant equipment, policies, procedures, and strategies to promote effective local, state, or national security operations for the protection of people, data, property, and institutions.

30. Law and Government (Minimal)

Knowledge of laws, legal codes, court procedures, precedents, government regulations, executive orders, agency rules, and the democratic political process.

31. No change.

32. Communications and Media (Minimal)

Knowledge of media production, communication, and dissemination techniques and methods. This includes alternative ways to inform and entertain via written, oral, and visual media.

33. Transportation (Minimal)

Knowledge of principles and methods for moving people or goods by air, rail, sea, or road, including the relative costs and benefits.

Original

Knowledge

Revision

34 - 38. Note that the revised Education and Training items, 1 - 5, were added to the end of the revised Knowledge Questionnaire used in the *O*NET Data Collection Program*.

39 - 54. Note that the revised Work Styles items, 1 - 16, were added to the end of the revised Knowledge Questionnaire used in the *O*NET Data Collection Program*.

Education and Training

Original

Part 1. LEVEL OF EDUCATION REQUIRED TO PERFORM THIS JOB

Please place an "X" next to the level of education required to perform this job. Note that this does not mean the education that you personally have obtained.

- a. **Less than a High School Diploma**
- b. **High School Diploma** (or High School Equivalence Certificate)
- c. **Post-Secondary Certificate** - awarded for training completed after high school (e.g., in Personal Services, Engineering-Related Technologies, Vocational Home Economics, Construction Trades, Mechanics and Repairers, Precision Production Trades)
- d. **Some College Courses**
- e. **Associate's Degree** (or other 2-year degree)
- f. **Bachelor's Degree**
- g. **Post-Baccalaureate Certificate** - awarded for completion of an organized program of study requiring 30 credit hours beyond the Bachelor's degree; designed for persons who have completed a Baccalaureate degree, but do not meet the requirements of academic degrees carrying the title of Master.
- h. **Master's Degree**
- i. **Post-Master's Certificate** - awarded for completion of an organized program of study of 60 credit hours beyond the Master's degree, but does not meet the requirements of academic degrees at the Doctoral level.
- j. **First Professional Degree** - awarded for completion of a program that:
 - ! requires at least 2 years of college work before entrance into the program,
 - ! includes a total of at least 6 academic years of college work to complete, and
 - ! provides all remaining academic requirements to begin practice in a profession
- k. **Doctoral Degree**
- l. **Post-Doctoral Certificate**

Revision

1. REQUIRED LEVEL OF EDUCATION

Please indicate the level of education required to perform this job. (Note that this does ***not*** mean the level of education that you personally have achieved.)

- G Less than a High School Diploma**
- G High School Diploma** (or GED or High School Equivalence Certificate)
- G Post-Secondary Certificate** - awarded for training completed after high school (e.g., in Personal Services, Engineering-Related Technologies, Vocational Home Economics, Construction Trades, Mechanics and Repairers, Precision Production Trades)
- G Some College Courses**
- G Associate's Degree** (or other 2-year degree)
- G Bachelor's Degree**
- G Post-Baccalaureate Certificate** - awarded for completion of an organized program of study requiring 30 credits beyond the Bachelor's degree; designed for people who have completed a Baccalaureate degree, but do not meet the requirements of academic degrees carrying the title of Master.
- G Master's Degree**
- G Post-Master's Certificate** - awarded for completion of an organized program of study requiring 60 credit hours beyond the Master's degree, but does not meet the requirements of academic degrees at the doctoral level.
- G First Professional Degree** - awarded for completion of a program that
 - ! requires at least 2 years of college work before entrance into the program,
 - ! includes a total of at least 6 academic years of work to complete, and
 - ! provides all remaining academic requirements to begin practice in a profession
- G Doctoral Degree**
- G Post-Doctoral Degree**

Education and Training

Original

Revision

Part 2. INSTRUCTIONAL PROGRAM REQUIRED

Delete.

Please indicate the instructional program that is required for this job by placing an "X" next to the appropriate area.

Part 3. LEVEL OF EDUCATION REQUIRED IN SPECIFIC SUBJECT AREAS

Delete.

Part 4. LICENSURE, CERTIFICATION, AND REGISTRATION

Delete.

Part 5. EXPERIENCE

If someone were being hired to perform this job, how much of the following would be required?

- Determine the level of experience that is required for this job. Use the following scale:

0 = not applicable or none

1 = up to and including 1 month

2 = over 1 month, up to and including 3 months

3 = over 3 months, up to and including 6 months

4 = over 6 months, up to and including 1 year

5 = over 1 year, up to and including 2 years

6 = over 2 years, up to and including 4 years

7 = over 4 years, up to and including 6 years

8 = over 6 years, up to and including 8 years

9 = over 8 years, up to and including 10 years

10 = over 10 years

- Place an "X" over the most appropriate value.

Education and Training

Original

1. **RELATED WORK EXPERIENCE**
(other jobs that prepared you for this job)

Revision

2. **If someone were being hired to perform this job, how much RELATED WORK EXPERIENCE would be required?** (That is, having other jobs that prepare the worker for the job)
 - G** None
 - G** Up to and including 1 month
 - G** Over 1 month, up to and including 3 months
 - G** Over 3 months, up to and including 6 months
 - G** Over 6 months, up to and including 1 year
 - G** Over 1 year, up to and including 2 years
 - G** Over 2 years, up to and including 4 years
 - G** Over 4 years, up to and including 6 years
 - G** Over 6 years, up to and including 8 years
 - G** Over 8 years, up to and including 10 years
 - G** Over 10 years
2. **ON-SITE OR IN-PLANT TRAINING**
(organized classroom study provided by the employer)
3. **If someone were being hired to perform this job, how much ON-SITE OR IN-PLANT TRAINING would be required?** (That is, organized classroom study provided by the employer)
 - G** None
 - G** Up to and including 1 month
 - G** Over 1 month, up to and including 3 months
 - G** Over 3 months, up to and including 6 months
 - G** Over 6 months, up to and including 1 year
 - G** Over 1 year, up to and including 2 years
 - G** Over 2 years, up to and including 4 years
 - G** Over 4 years, up to and including 6 years
 - G** Over 6 years, up to and including 8 years
 - G** Over 8 years, up to and including 10 years
 - G** Over 10 years

Education and Training

Original

3. **ON-THE-JOB TRAINING (OJT)**
(serving as a learner or trainee on the job under the instruction of a more experienced worker)

4. **APPRENTICESHIP**
(having served in a registered US Department of Labor apprenticeship program and received a certificate of completion)

Revision

4. **If someone were being hired to perform this job, how much ON-THE-JOB TRAINING would be required?** (That is, serving as a learner or trainee on the job under instruction of a more experienced worker)

- G** None
- G** Up to and including 1 month
- G** Over 1 month, up to and including 3 months
- G** Over 3 months, up to and including 6 months
- G** Over 6 months, up to and including 1 year
- G** Over 1 year, up to and including 2 years
- G** Over 2 years, up to and including 4 years
- G** Over 4 years, up to and including 6 years
- G** Over 6 years, up to and including 8 years
- G** Over 8 years, up to and including 10 years
- G** Over 10 years

5. **If someone were being hired to perform this job, how much APPRENTICESHIP would be required?** (That is, having served in a registered US Department of Labor apprenticeship program and received a certificate of completion)

- G** None
- G** Up to and including 1 month
- G** Over 1 month, up to and including 3 months
- G** Over 3 months, up to and including 6 months
- G** Over 6 months, up to and including 1 year
- G** Over 1 year, up to and including 2 years
- G** Over 2 years, up to and including 4 years
- G** Over 4 years, up to and including 6 years
- G** Over 6 years, up to and including 8 years
- G** Over 8 years, up to and including 10 years
- G** Over 10 years

Work Styles

Original

Revision

- | | |
|--|--|
| <p>1. Achievement/Effort
Job requires establishing and maintaining personally challenging achievement goals and exerting effort toward mastering tasks.</p> | <p>1. No change.</p> |
| <p>2. Persistence
Job requires persistence in the face of obstacles on the job.</p> | <p>2. Persistence (Minimal)
Job requires persistence in the face of obstacles.</p> |
| <p>3. Initiative
Job requires being willing to take on responsibilities and challenges.</p> | <p>3. Initiative (Minimal)
Job requires a willingness to take on responsibilities and challenges.</p> |
| <p>4. Energy
Job requires the energy and stamina to accomplish work tasks.</p> | <p>Delete</p> |
| <p>5. Leadership Orientation
Job requires a willingness to lead, take charge and offer opinions and direction.</p> | <p>4. Leadership (Minimal)
Job requires a willingness to lead, take charge, and offer opinions and direction.</p> |
| <p>6. Cooperation
Job requires being pleasant with others on the job and displaying a good-natured, cooperative attitude.</p> | <p>5. No change.</p> |
| <p>7. Concern for Others
Job requires being sensitive to others' needs and feelings and being understanding and helpful to others on the job.</p> | <p>6. No change.</p> |
| <p>8. Social Orientation
Job requires preferring to work with others rather than alone, and being personally connected with others on the job.</p> | <p>7. No change.</p> |
| <p>9. Self-Control
Job requires maintaining composure, keeping emotions in check, controlling anger, and avoiding aggressive behavior, even in very difficult situations.</p> | <p>8. No change.</p> |

Work Styles

<u>Original</u>	<u>Revision</u>
10. Stress Tolerance Job requires accepting criticism and dealing calmly and effectively with high-stress situations.	9. No change.
11. Adaptability/Flexibility Job requires being open to change (positive or negative) and to considerable variety in the workplace.	10. No change.
12. Dependability Job requires being reliable, responsible, and dependable, and fulfilling obligations.	11. No change.
13. Attention to Detail Job requires being careful about details and thorough in completing tasks.	12. No change.
14. Integrity Job requires being honest and avoiding unethical behavior.	13. Integrity (Minimal) Job requires being honest and ethical.
15. Independence Job requires developing one's own ways of doing things, guiding oneself with little or no supervision, and depending on oneself to get things done.	14. No change.
16. Innovation Job requires creativity and alternative thinking to come up with new ideas for and answers to work-related problems.	15. Innovation (Minimal) Job requires creativity and alternative thinking to develop new ideas for and answers to work-related problems.
17. Analytical Thinking Job requires analyzing information, and using logic to address work or job issues and problems.	16. Analytical Thinking (Minimal) Job requires analyzing information and using logic to address work-related issues and problems.

Skills

Original

1. **Reading Comprehension**
Understanding written sentences and paragraphs in work-related documents.
2. **Active Listening**
Listening to what other people are saying and asking questions as appropriate.
3. **Writing**
Communicating effectively with others in writing as indicated by the needs of the audience.
4. **Speaking**
Talking to others to effectively convey information.
5. **Mathematics**
Using mathematics to solve problems.
6. **Science**
Using scientific methods to solve problems.
7. **Critical Thinking**
Using logic and analysis to identify the strengths and weaknesses of different approaches.
8. **Active Learning**
Working with new material or information to grasp its implications.

Revision

1. No change.
2. **Active Listening (Moderate)**
Giving full attention to what other people are saying, taking time to understand the points being made, asking questions as appropriate, and not interrupting at inappropriate times.
3. **Writing (Minimal)**
Communicating effectively in writing as appropriate for the needs of the audience.
4. **Speaking (Minimal)**
Talking to others to convey information effectively.
5. No change.
6. **Science (Minimal)**
Using scientific rules and methods to solve problems.
7. **Critical Thinking (Minimal)**
Using logic and reasoning to identify the strengths and weaknesses of alternative solutions, conclusions or approaches to problems.
8. **Active Learning (Moderate)**
Understanding the implications of new information for both current and future problem-solving and decision-making.

The modified definition is more congruent with listening skill as it is defined in the interpersonal skills literature.

Skills

Original

- 9. Learning Strategies**
Using multiple approaches when learning or teaching new things.
- 10. Monitoring**
Assessing how well one is doing when learning or doing something.
- 11. Social Perceptiveness**
Being aware of others' reactions and understanding why they react the way they do.
- 12. Coordination**
Adjusting actions in relation to others' actions.
- 13. Persuasion**
Persuading others to approach things differently.
- 14. Negotiation**
Bringing others together and trying to reconcile differences.
- 15. Instructing**
Teaching others how to do something.
- 16. Service Orientation**
Actively looking for ways to help people.

Revision

- 9. Learning Strategies (Moderate)**
Selecting and using training/instructional methods and procedures appropriate for the situation when learning or teaching new things.
- 10. Monitoring (Moderate)**
Monitoring/Assessing performance of yourself, other individuals, or organizations to make improvements or take corrective action.
- 11. Social Perceptiveness (Minimal)**
Being aware of others' reactions and understanding why they react as they do.
- 12.** No change.
- 13. Persuasion (Minimal)**
Persuading others to change their minds or behavior.
- 14.** No change.
- 15.** No change.
- 16.** No change.

Skills

Original

- 17. Problem Identification**
Identifying the nature of problems.
- 18. Information Gathering**
Knowing how to find information and identifying essential information.
- 19. Information Organization**
Finding ways to structure or classify multiple pieces of information.
- 20. Synthesis/Reorganization**
Reorganizing information to get a better approach to problems or tasks.
- 21. Idea Generation**
Generating a number of different approaches to problems.
- 22. Idea Evaluation**
Evaluating the likely success of an idea in reaction to the demands of the situation.
- 23. Implementation Planning**
Developing approaches for implementing an idea.
- 24. Solution Appraisal**
Observing and evaluating the outcomes of a problem solution to identify lessons learned or redirect efforts.
- 25. Operations Analysis**
Analyzing needs and product requirements to create a design.
- 26. Technology Design**
Generating or adapting equipment and technology to serve user needs.
- 27. Equipment Selection**
Determining the kind of tools and equipment needed to do a job.
- 28. Installation**
Installing equipment, machines, wiring, or programs to meet specifications.
- 29. Programming**
Writing computer programs for various purposes.

Revision

- 17. Complex Problem Solving**
(Combination of original items 17-24)
Identifying complex problems and reviewing related information to develop and evaluate options and implement solutions.

- | | |
|------------|------------|
| 18. | No change. |
| 19. | No change. |
| 20. | No change. |
| 21. | No change. |
| 22. | No change. |

Skills

Original

30. Testing

Conducting tests to determine whether equipment, software, or procedures are operating as expected.

33. Product Inspection

Inspecting and evaluating the quality of products.

31. Operation Monitoring

Watching gauges, dials, or other indicators to make sure a machine is working properly.

32. Operation and Control

Controlling operations of equipment or systems.

34. Equipment Maintenance

Performing routine maintenance and determining when and what kind of maintenance is needed.

35. Troubleshooting

Determining what is causing an operating error and deciding what to do about it.

36. Repairing

Repairing machines or systems using the needed tools.

37. Visioning

Developing an image of how a system should work under ideal conditions.

38. Systems Perceptions

Determining when important changes have occurred in a system or are likely to occur.

39. Identification of Downstream Consequences

Determining the long-term outcomes of a change in operations.

Revision

23. Quality Control Analysis (Combination of original items 30 and 33)

Conducting tests and inspections of products, services, or processes to evaluate quality or performance.

24. Operations Monitoring (Minimal)

Watching gauges, dials, or other indicators to make sure a machine is working properly.

25. No change.

26. Equipment Maintenance (Minimal)

Performing routine maintenance on equipment and determining when and what kind of maintenance is needed.

27. Troubleshooting (Minimal)

Determining causes of operating errors and deciding what to do about it.

28. No change.

29. Systems Analysis (Combination of original items 37-39)

Determining how a system should work and how changes in conditions, operations, and the environment will affect outcomes.

Skills

Original

- 40. Identification of Key Causes**
Identifying the things that must be changed to achieve a goal.
- 42. Systems Evaluation**
Looking at many indicators of system performance, taking into account their accuracy.
- 41. Judgment and Decision Making**
Weighing the relative costs and benefits of a potential action.
- 43. Time Management**
Managing one's own time and the time of others.
- 44. Management of Financial Resources**
Determining how money will be spent to get the work done, and accounting for these expenditures.
- 45. Management of Material Resources**
Obtaining and seeing to the appropriate use of equipment, facilities, and materials needed to do certain work.
- 46. Management of Personnel Resources**
Motivating, developing, and directing people as they work, identifying the best people for the job.

Revision

- 30. Systems Evaluation** (Combination of original items 40 and 42)
Identifying measures or indicators of system performance and the actions needed to improve or correct performance, relative to the goals of the system.
- 31. Judgment and Decision Making**
(Minimal)
Considering the relative costs and benefits of potential actions to choose the most appropriate one.
- 32.** No change.
- 33.** No change.
- 34.** No change.
- 35.** No change.

Work Context

Original

Revision

- | | |
|--|---|
| 1. Formality of Communication
How formal is most of the job-related information that the worker gives and receives on this job? | Delete |
| 2. Communication Methods.
How frequently does the job require the use of the following communication methods? | |
| a. Face-to-Face (individuals). | a. (Combined with b in Revised item 1.) |
| b. Face-to-Face (group/team meetings). | 1. How often does <i>your current job</i> require <u>face-to-face discussions with individuals and within teams</u>? |
| c. Public Speaking. | 2. How frequently does <i>your current job</i> require <u>public speaking</u>? (one speaker with an audience) |
| d. Video Conference. | Delete. |
| e. Voice Mail. | Delete. |
| f. Telephone. | 3. How frequently does <i>your current job</i> require <u>telephone conversation</u>? |
| g. Interactive (same-time) computer communication. | Delete. |
| h. Electronic Mail. | 4. How frequently does <i>your current job</i> require <u>electronic mail</u>? |
| i. Handwritten Notes of Messages. | Delete. |
| j. Letters and Memos. | 5. How frequently does <i>your current job</i> require <u>written letters and memos</u>? |
| k. Written Reports. | Delete. |
| 3. Objectivity vs. Subjectivity of Information Communicated.
How objective or subjective is the information communicated in this job? | Delete. |

Work Context

Original

4. Job Required Social Interaction.
How much does this job require the worker to be in contact (face-to-face, by telephone, or otherwise) with others in order to perform it?
5. Privacy of Communications.
To what extent can an individual's work materials and communications (face-to-face, phone, fax, E-mail, etc.) be monitored by others?
6. Job Interactions.

How important are interactions requiring the worker to:
 - a. Supervise, coach, train, or develop other employees?
 - b. Persuade someone to a course of action (informally) or influence others to buy something (to sell)?
 - c. Provide a service to others (e.g., customers)?
 - d. Take a position opposed to coworkers or others?
 - e. Work with or contribute to a work group or team to perform this job?
 - f. Deal with external customers (e.g., retail sales) or the public in general (e.g., police work)?

Revision

6. How much contact with others (by telephone, face-to-face, or otherwise) is required to perform *your current job*?
- Delete.
- Delete.
- Delete.
- Delete.
7. How important are interactions that require you to work with or contribute to a work group or team to perform *your current job*?
8. In *your current job*, how important are interactions that require you to deal with external customers (as in retail sales) or the public in general (as in police work)?

Work Context

Original

- g. Coordinate or lead others in accomplishing work activities (not supervision)?
- 7. Responsibility for Others' Health and Safety. How responsible is the worker for others' health and safety on this job?
- 8. Responsibility for Work Outcomes and Results. How responsible is the worker for work outcomes and results of other workers?
- 9. How frequently do the job requirements place the worker in conflict situations?
- 10. How frequently does the worker have to deal with unpleasant, angry, or discourteous individuals as part of the job requirements?
- 11. How frequently does this job require the worker to deal with physical aggression of violent individuals?
- 12. Work Setting. How frequently does this job require the worker to work:
 - a. Indoors, environmentally controlled?
 - b. Indoors, not environmentally controlled (e.g., warehouse without air conditioning)?
 - c. Outdoors, exposed to all weather conditions?

Revision

- 9. In *your current job*, how important are interactions that require you to coordinate or lead others in accomplishing work activities? (*not as a supervisor or team leader*)
- 10. How responsible are you for the health and safety of other workers on *your current job*?
- 11. How responsible are you for work outcomes and results of other workers on *your current job*?
- 12. How often are conflict situations a part of *your current job*?
- 13. How often is dealing with unpleasant, angry or discourteous people a part of *your current job*?
- 14. How often is dealing with violent or physically aggressive people a part of *your current job*?
- 15. How often does *your current job* require you to work indoors in an environmentally controlled environment (like a warehouse with air conditioning)?
- 16. How often does *your current job* require you to work in an environment that is not environmentally controlled (like a warehouse without air conditioning)?
- 17. How often does *your current job* require you to work outdoors, exposed to all weather conditions?

Work Context

Original

- d. Outdoors, under cover (e.g., open shed)?
 - e. In an open vehicle or operating open equipment (e.g., tractor)?
 - f. In an enclosed vehicle or operating enclosed equipment (e.g., automobile)?
13. Privacy of Work Area.
How private is the work area for this job?
14. Physical Proximity.
To what extent does this job require the worker to perform job tasks in close physical proximity to other people?
15. Environmental Conditions.
How often during a usual work period is the worker exposed to the following conditions:
- a. Sounds and noise levels that are distracting and uncomfortable?
 - b. Very hot (above 90° F) or very cold (under 32° F) temperatures?
 - c. Extremely bright or inadequate lighting conditions?
 - d. Contaminants (pollutants, gases, dust, odors, etc.)?
 - e. Cramped work space that requires getting into awkward positions?

Revision

18. How often does *your current job* require you to work outdoors, under cover (like in an open shed)?
19. How often does *your current job* require you to work in an open vehicle or operating equipment (like a tractor)?
20. How often does *your current job* require you to work in a closed vehicle or operate enclosed equipment (like a car)?
- Delete.
21. How physically close to other people are you when you perform *your current job*?
22. In *your current job*, how often are you exposed to sounds and noise levels that are distracting and uncomfortable?
23. In *your current job*, how often are you exposed to very hot (above 90° F) or very cold (under 32° F) temperatures?
24. In *your current job*, how often are you exposed to extremely bright or inadequate lighting conditions?
25. In *your current job*, how often are you exposed to contaminants (such as pollutants, gases, dust or odors)?
26. In *your current job*, how often are you exposed to cramped work space that requires getting into awkward positions?

Work Context

Original

- f. Whole body vibration (e.g., operating a jackhammer or earth-moving equipment)?

16. Exposure to Radiation.

- a. Frequency. How often does this job require the worker to be exposed to radiation?
- b. Likelihood of Injury. What is the likelihood that the worker would be injured as a result of being exposed to radiation while performing this job?
- c. Degree of Injury. If injury, due to exposure to radiation, were to occur while performing this job, how serious would be the likely outcome?

17. Exposure to Diseases/Infections (e.g., patient care, some laboratory work, sanitation control, etc.)

- a. Frequency. How often does this job require the worker to be exposed to diseases/infections?
- b. Likelihood of Injury. What is the likelihood that the worker would be injured as a result of being exposed to diseases/infections while performing this job?
- c. Degree of Injury. If injury, due to exposure to diseases/infections, were to occur while performing this job, how serious would be the likely outcome?

Revision

27. **In your *current job*, how often are you exposed to whole body vibration (like operating a jackhammer or earth moving equipment)?**

28. **How often does your *current job* require that you be exposed to radiation?**

Delete.

Delete.

29. **How often does your *current job* require that you be exposed to diseases or infection? This can happen with workers in patient care, some laboratory work, sanitation control, etc.**

Delete.

Delete.

Work Context

Original

Revision

18. Exposure to High Places (e.g., heights above 8 feet on ladders, poles, scaffolding, catwalks, etc.)

a. Frequency. How often does this job require the worker to be exposed to high places?

b. Likelihood of Injury. What is the likelihood that the worker would be injured as a result of being exposed to high places while performing this job?

c. Degree of Injury. If injury, due to exposure to high places, were to occur while performing this job, how serious would be the likely outcome?

19. Exposure to Hazardous Conditions (e.g., high voltage electricity, combustibles, explosives, chemicals; do not include hazardous equipment or situations - see questions 20 and 21).

a. Frequency. How often does this job require the worker to be exposed to hazardous conditions?

b. Likelihood of Injury. What is the likelihood that the worker would be injured as a result of being exposed to hazardous conditions while performing this job?

c. Degree of Injury. If injury, due to exposure to hazardous conditions, were to occur while performing this job, how serious would be the likely outcome?

30. How often does *your current job* require that you be exposed to high places? This can happen for workers who work on poles, scaffolding, catwalks, or ladders longer than 8 feet in length.

Delete.

Delete.

31. How often does *your current job* require that you be exposed to hazardous conditions? This can happen when working with high voltage electricity, flammable material, explosives, or chemicals. Do not include working with hazardous equipment.

Delete.

Delete.

Work Context

Original

Revision

20. Exposure to Hazardous Equipment (e.g., saws, machinery/mechanical parts; include exposure to vehicular traffic, but not driving a vehicle).

a. Frequency. How often does this job require the worker to be exposed to hazardous equipment?

b. Likelihood of Injury. What is the likelihood that the worker would be injured as a result of being exposed to hazardous equipment while performing this job?

c. Degree of Injury. If injury, due to exposure to hazardous equipment, were to occur while performing this job, how serious would be the likely outcome?

21. Exposure to Hazardous Situations involving likely cuts, bites, stings, or minor burns

a. Frequency. How often does this job require the worker to be exposed to hazardous situations?

b. Likelihood of Injury. What is the likelihood that the worker would be injured as a result of being exposed to hazardous situations while performing this job?

c. Degree of Injury. If injury, due to exposure to hazardous situations, were to occur while performing this job, how serious would be the likely outcome?

22. Body Positioning.
How much time in a usual work period does the worker spend:

32. How often does *your current job* require that you be exposed to hazardous equipment? This includes working with saws, close to machinery with exposed moving parts, or working near vehicular traffic (but not including driving a vehicle).

Delete.

Delete.

33. How often does *your current job* require that you be exposed to minor burns, cuts, bites, or stings?

Delete.

Delete.

Work Context

Original

Revision

- | | |
|---|---|
| a. Sitting? | 34. How much time in <i>your current job</i> do you spend <u>sitting</u> ? |
| b. Standing? | 35. How much time in <i>your current job</i> do you spend <u>standing</u> ? |
| c. Climbing ladders, scaffolds, poles, etc.? | 36. How much time in <i>your current job</i> do you spend <u>climbing ladders, scaffolds, poles, etc.</u> ? |
| d. Walking or running? | 37. How much time in <i>your current job</i> do you spend <u>walking or running</u> ? |
| e. Kneeling, crouching, stooping, or crawling? | 38. How much time in <i>your current job</i> do you spend <u>kneeling, crouching, stooping, or crawling</u> ? |
| f. Keeping or regaining balance? | 39. How much time in <i>your current job</i> do you spend <u>keeping or regaining your balance</u> ? |
| g. Using hands to handle, control, or feel objects, tools, or controls? | 40. How much time in <i>your current job</i> do you spend <u>using your hands to handle, control, or feel objects, tools, or controls</u> ? |
| h. Bending or twisting the body? | 41. How much time in <i>your current job</i> do you spend <u>bending or twisting your body</u> ? |
| i. Making repetitive motions? | 42. How much time in <i>your current job</i> do you spend <u>making repetitive motions</u> ? |
| 23. Work Attire. | |
| How often does the worker wear: | |
| a. Business clothes, such as neckties and dresses that are often worn in offices? | Delete. |
| b. A special uniform, such as that of a commercial pilot, nurse, police officer, or military personnel? | Delete. |
| c. Work clothing such as that worn by production or maintenance workers? | Delete. |

Work Context

Original

- d. Common protective or safety attire, such as safety shoes, glasses, gloves, hearing protection, hard-hat, or personal flotation device?
 - e. Specialized protective or safety attire, such as a breathing apparatus, safety harness, full protection suit, or radiation protection?
24. Consequences of Error.
How serious would the result usually be if the worker made a mistake that was not readily correctable?
25. Impact of Decisions.
- a. Level
What results do the worker's decision usually have on other people, the financial resources, and/or the image or reputation of the organization?
 - b. Frequency
How frequently is the worker required to make decisions that affect other people, the financial resources, and/or the image and reputation of the organization?
26. Responsibility/Accountability.
To what extent is this job assigned accountability for final work outcomes or results?
27. Decision Latitude.
Indicate the amount of freedom the worker has to make decisions without supervision.

Revision

43. In *your current job*, how often do you wear common protective or safety equipment such as safety shoes, glasses, gloves, hearing protection, hard hats, or life jackets?
44. In *your current job*, how often do you wear specialized protective or safety equipment, such as breathing apparatus, safety harness, full protection suits, or radiation protection?
45. How serious a mistake can you make on *your current job* (one you can't easily correct)?
46. In *your current job*, what results do your decisions usually have on other people or the image or reputation or financial resources of your employer?
47. In *your current job*, how often do your decisions affect other people or the image or reputation or financial resources of your employer?
- Delete.
48. In *your current job*, how much freedom do you have to make decisions without supervision?

Work Context

Original

Revision

- | | |
|---|---|
| 28. Frustrating Circumstances.
To what extent do frustrating circumstances (“road blocks” to work that are beyond the worker’s control) hinder the accomplishment of this job? | Delete. |
| 29. Degree of Automation.
Indicate the level of automation of this job. | 49. How <u>automated</u> is your current job? |
| 30. Task Clarity.
To what extent is the worker clear about what is to be done on this job and how work performance is to be evaluated? | Delete. |
| 31. How important is being very exact or highly accurate in performing this job? | 50. How important to your current job is being <u>very exact or highly accurate</u>? |
| 32. How important is it to be sure that all the details of this job are performed and everything is done completely? | Delete. |
| 33. How important is being constantly aware of either frequently changing events (e.g., security guard watching for shoplifters) or infrequent events (e.g., radar operator watching for tornadoes) to performing this job? | Delete. |
| 34. How important is repeating the same physical activities (e.g., key entry) or mental activities (e.g., checking entries in a ledger) over and over, without stopping, to performing this job? | 51. How important to your current job are <u>continuous, repetitious physical activities</u> (like key entry) or <u>mental activities</u> (like checking entries in a ledger)? |
| 35. Structured vs. Unstructured Work.
To what extent is this job structured for the worker, rather than allowing the worker to determine tasks, priorities, and goals? | 52. How much freedom do you have to <u>determine the tasks, priorities or goals of your current job</u>? |
| 36. Level of Competition.
To what extent does this job require the worker to compete or to be aware of competitive pressures? | 53. How <u>competitive</u> is your current job? |

Work Context

Original

37. Time Pressure.
Frequency. How often does this job require the worker to meet strict deadlines?
38. How important is working under frequent distractions or interruptions to performing this job?
39. How important is it to this job that the pace is determined by the speed of equipment or machinery? (This does not refer to keeping busy at all times on this job.)
40. Please check the usual work schedule for this job. (Check only one.)
- Regular Work (established routine, with set schedule)
 - Irregular Work (subject to weather conditions, production demands, contract duration)
 - Seasonal Basis (only work during certain times of the year)
41. Please check the usual work shift for this job. (Check only one.)
- Day Shift
 - Other than Day Shift (i.e., evening shift or night shift)
 - Split or Variable Shift (work busy times or shift changes due to staffing demands)
 - Rotating Shift (rotate days, evenings, nights)
42. Please check the usual work shift duration. (Check only one.)
- Paid for less than 8 hours
 - Paid for 8 hours
 - Paid for more than 8 hours

Revision

54. How often does *your current job* require you to meet strict deadlines?
- Delete.
55. How important to *your current job* is keeping a pace set by machinery or equipment?
56. How regular is your work schedule on *your current job*?

Delete.

Delete.

Work Context

Original

Revision

43. Please check the usual overtime work.
(Check all that apply.)
- a. None
 - b. Overtime at request of employer
 - c. Available for call-in to work
 - d. Weekend work (not part of regular shift)

Delete.

44. Please check the number of hours typically worked in one week. (Check only one.)
- a. Less than 40 hours
 - b. 40 hours
 - c. More than 40 hours

57. How many hours do you work in a typical week on *your current job*?

45. Please check the usual work cycle for this job. (Check only one.)
- a. 1 week
 - b. 2 weeks
 - c. 1 month (28 or more days)

Delete.

46. Please check the number of days usually worked in the cycle. (Check only one.)
- a. Less than 3
 - b. 3
 - c. 4
 - d. 5
 - e. 6
 - f. 7
 - g. 8-10
 - h. 11-14
 - i. 15-20
 - j. More than 20

Delete.

Note: For the revised Work Context items that were not deleted, all changes that have occurred were slight wording and formatting changes and are considered minimal.